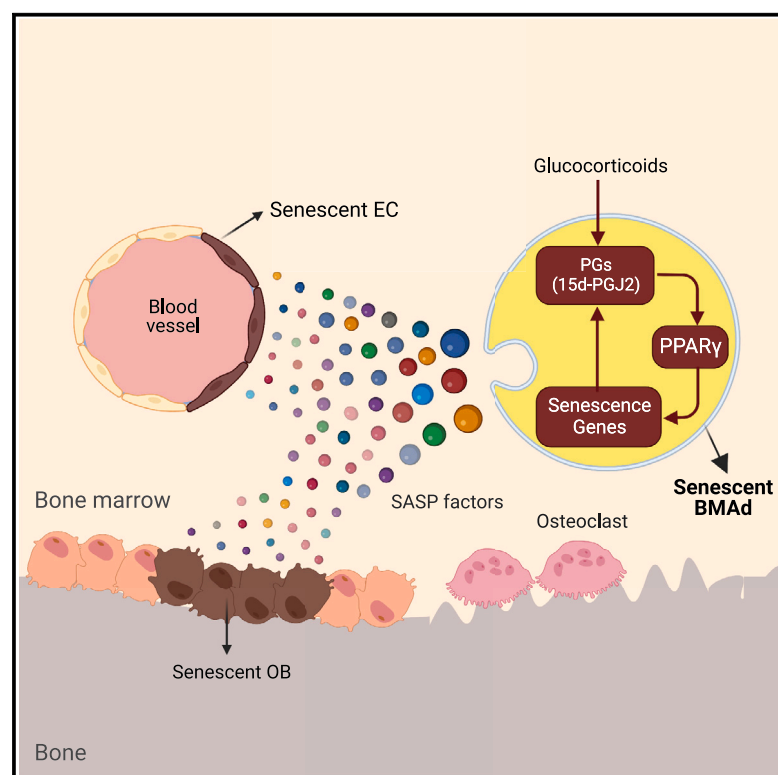


Cell Metabolism

Oxylipin-PPAR γ -initiated adipocyte senescence propagates secondary senescence in the bone marrow

Graphical abstract



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In brief

Liu et al. reveal that glucocorticoids directly induce cellular senescence of bone-marrow adipocytes (BMAd) through a prostaglandin-PPAR γ -INK positive feedback loop. The senescent BMAd triggers a secondary senescence in the surrounding bone cells by producing senescence-associated secretory phenotype (SASP) factors, leading to bone deterioration.

Highlights

- BMAd undergo rapid senescence after glucocorticoid treatment
- Glucocorticoids activate prostaglandin-PPAR γ -INK signaling to initiate BMAd senescence
- Senescent BMAd acquire an SASP, causing secondary senescence of surrounding bone
- Blockage of BMAd senescence inhibits the secondary senescence, rescuing bone loss



Article

Oxylipin-PPAR γ -initiated adipocyte senescence propagates secondary senescence in the bone marrow

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<https://doi.org/10.1016/j.cmet.2023.03.005>

SUMMARY

The chronic use of glucocorticoids decreases bone mass and quality and increases bone-marrow adiposity, but the underlying mechanisms remain unclear. Here, we show that bone-marrow adipocyte (BMA) lineage cells in adult mice undergo rapid cellular senescence upon glucocorticoid treatment. The senescent BMAs acquire a senescence-associated secretory phenotype, which spreads senescence in bone and bone marrow. Mechanistically, glucocorticoids increase the synthesis of oxylipins, such as 15d-PGJ2, for peroxisome proliferator-activated receptor gamma (PPAR γ) activation. PPAR γ stimulates the expression of key senescence genes and also promotes oxylipin synthesis in BMAs, forming a positive feedback loop. Transplanting senescent BMAs into the bone marrow of healthy mice is sufficient to induce the secondary spread of senescent cells and bone-loss phenotypes, whereas transplanting BMAs harboring a *p16INK4a* deletion did not show such effects. Thus, glucocorticoid treatment induces a lipid metabolic circuit that robustly triggers the senescence of BMA lineage cells that, in turn, act as the mediators of glucocorticoid-induced bone deterioration.

INTRODUCTION

Bone-marrow adipose tissue mass increases under diverse physiological and clinical conditions, including aging, obesity, type 2 diabetes, anorexia, hyperlipidemia, and osteoporosis, as well as in response to radiotherapy and treatment with thiazolidinediones (TZDs) and glucocorticoids.¹ Many of these conditions are associated with decreased bone mass and increased fracture risk. It is now accepted that the increase in the number of bone-marrow adipocytes (BMAs) in the above contexts regulate the activities of other cell populations within and outside the bone by secreting cytokines, chemokines, lipid species, and adipokines.^{2–5} However, these studies have been conducted primarily *in vitro*.⁴ It remains unknown whether BMAs have a negative impact on bone remodeling via secretion of these factors *in vivo* under pathological conditions. The mechanisms by which BMAs acquire the secretory phenotype are also unclear.

Chronic glucocorticoid treatment is a common cause of secondary osteoporosis, leading to increased fracture risk.^{6–8} Glucocorticoids exert direct effects on many bone and bone-marrow cells that express the nuclear glucocorticoid receptor, including osteoclasts,^{8–14} mesenchymal stem/progenitor cells,^{15,16} osteoblasts (OBs),¹⁷ osteocytes,^{18–20} and adipocytes.^{21,22} Pharmacological doses of glucocorticoids impair osteogenesis by promoting OB and osteocyte apoptosis⁹ and by inhibiting OB differentia-

tion.¹⁷ Glucocorticoid treatment also increases osteoclast bone resorption by upregulating receptor activator of nuclear factor kappa-B ligand (RANKL) expression and promoting the survival of osteoclasts.^{23–25} Glucocorticoids inhibit angiogenesis, and thus osteogenesis, by decreasing endothelial growth factors, such as vascular-endothelial growth factor (VEGF), dimeric platelet-derived growth factor B (PDGF-BB), and angiogenin (ANG).^{26–28} Some of the adverse effects of glucocorticoids on bone may be mediated through a change in systemic metabolism because glucocorticoids have a major impact on glucose and lipid metabolism in the body.^{29,30} In addition, increased concentrations of glucocorticoids trigger adipogenesis both *in vitro* and *in vivo* by upregulating the expression of *Pparg2* or *Cebpa*.³¹ In the long bones of glucocorticoid-treated animals, bone-marrow adiposity accrual is concomitant with (or even precedes) a decrease in bone-mineral density, trabecular bone, or OB numbers.^{32,33} The fate change of BMAs and their role in glucocorticoid-induced osteoporosis remain poorly understood.

Cellular senescence refers to a stable cell-cycle arrest process, which is mediated mainly by the activation of p53-p21- or p16-Rb-regulated pathways.^{34,35} In addition, senescent cells are metabolically active and develop other characteristics, including enlarged and flattened morphology, increased senescence-associated β -galactosidase (SA- β Gal), loss of high mobility group box 1 (HMGB1), downregulation of lamin-B1,



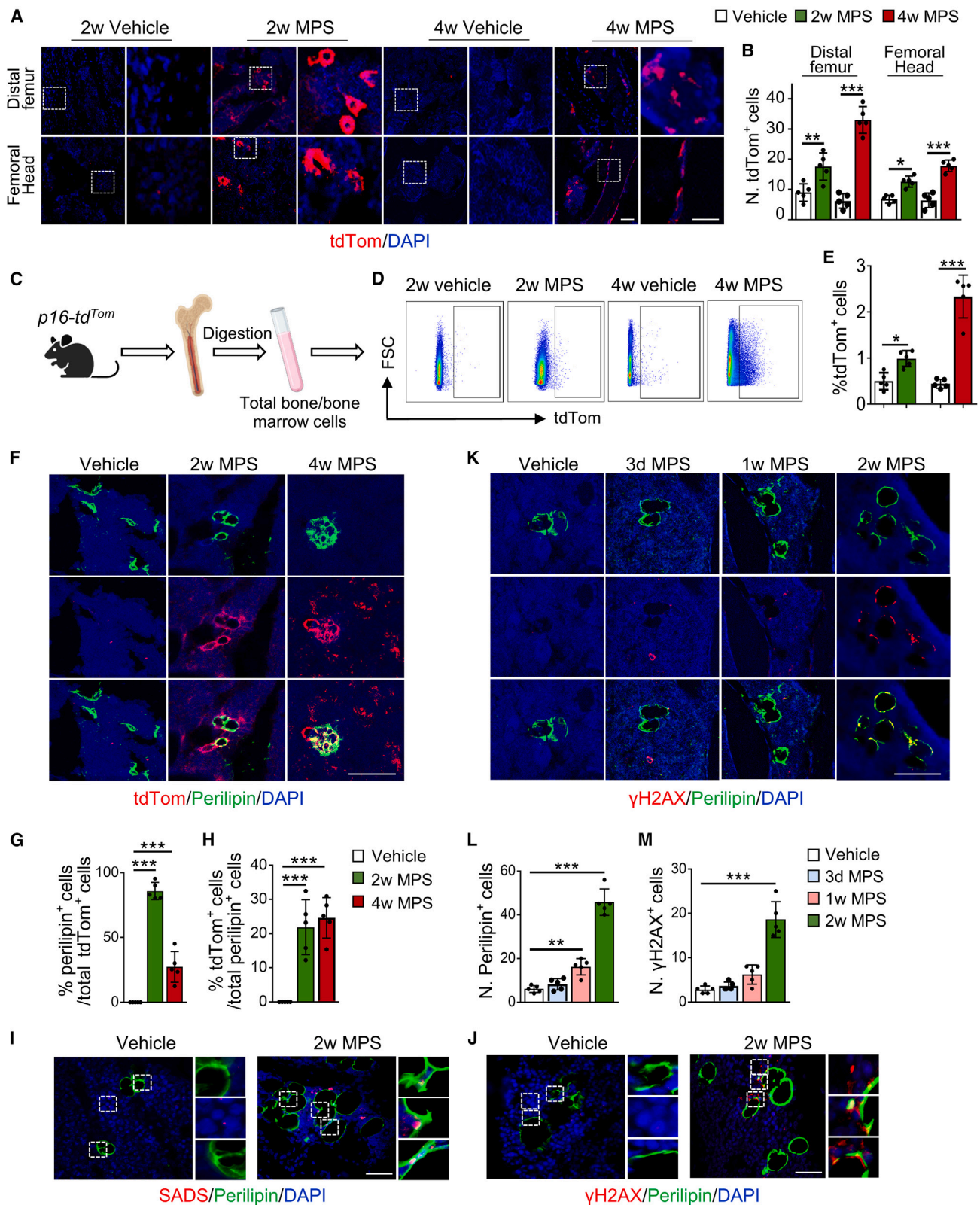


Figure 1. Glucocorticoid treatment induces rapid senescence of BMADs in mice

3-month-old *p16^{tdTom}* mice were treated with methylprednisolone (MPS) at 10 mg/m²/day or vehicle by daily intraperitoneal injection for different time periods as indicated.

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formation of senescence-associated distension of satellites (SADSs), and most importantly development of the senescence-associated secretory phenotype (SASP).^{34–42} The SASP factors include cytokines, chemokines, growth factors, proteases, and extracellular components that can affect biological signaling in an autocrine or paracrine manner.^{43,44} Recently, nonprotein components in the SASP have also been found. For example, senescent cells secrete oxylipins, especially prostaglandin-related lipids, and the prostaglandins reinforce the senescence phenotype and SASP secretion.⁴⁵ SASP factors have been shown to spread senescence to neighboring non-senescent cells through a paracrine manner or cell-cell contact.⁴⁶ We recently identified that growing young mice accumulate vascular-endothelial cell senescence in the metaphysis of long bones in response to glucocorticoid treatment.²⁷ It remains unclear whether glucocorticoid treatment also induces cellular senescence in adult mice. Here, we show that, unlike bone vascular cell senescence in primary spongiosa of growing young mice, glucocorticoid treatment induces rapid cellular senescence in BMAd lineage cells. The senescence program is initiated in lineage-committed adipocyte precursors and progresses into full cellular senescence in mature BMAds, which secrete a panel of SASP factors to drive secondary senescence within the bone environment. Using *in vitro* and *in vivo* approaches, we also show that peroxisome proliferator-activated receptor gamma (PPAR γ) and oxylipins (e.g., prostaglandin D2 derivatives) form a vicious positive feedback loop to trigger and maintain glucocorticoid-induced BMAd senescence.

RESULTS

Glucocorticoid treatment induces rapid BMAd senescence in mice

We previously demonstrated that glucocorticoid treatment in growing young mice induces vascular-endothelial cell senescence in the metaphysis of long bone.²⁷ To examine whether glucocorticoid treatment also induces cellular senescence in the adult skeleton, we treated 3-month-old C57BL/6 mice with methylprednisolone (MPS), a commonly used synthetic glucocorticoid, for different durations. MPS treatment induced a progressive increase in bone-marrow adipose tissue mass, as evidenced by the higher percentage of bone-marrow adipose tissue volume per total volume (%AV/TV) in decalcified distal

femurs stained with osmium tetroxide (OsO₄) in MPS-treated mice compared with vehicle-treated mice (Figures S1A and S1B). MPS treatment for 2 weeks induced an increase in the number of SA- β Gal⁺ cells in both the distal femur and femoral head (Figures S1C and S1D). However, MPS treatment for 4 weeks resulted in many more SA- β Gal⁺ cells in both bone regions compared with 2-week MPS treatment or vehicle treatment (Figures S1C and S1D). Moreover, the number of cells positive for HMGB1 (the loss of nuclear expression of which is another feature of senescent cells) was moderately lower in bone tissue of 2-week MPS-treated mice relative to vehicle-treated mice (Figures S1E and S1F). However, the lower incidence of HMGB1⁺ cells in both the distal femur and femoral head was much greater in 4-week MPS-treated mice compared with vehicle-treated mice (Figures S1E and S1F). We also used a mouse-senescence reporter strain harboring *p16^{tdTom}*, in which p16INK4a-activated cells (tdTom⁺) can be detected.⁴⁷ *In situ* fluorescence analysis of the femoral bone-tissue sections showed that MPS treatment induced a greater number of tdTom⁺ cells in both the distal femur and femoral head in a time-dependent manner (Figures 1A and 1B). Consistent with these results, we also detected a progressive accumulation of tdTom⁺ senescent cells in the femoral bone in response to MPS treatment by flow cytometry analysis (Figures 1C–1E). Therefore, glucocorticoid treatment induces a gradual increase in senescent cells in the adult long bone.

We noticed that most of the tdTom⁺ cells at 2 weeks after MPS treatment were large, ring-like cells that were like adipocytes in appearance. Indeed, immunofluorescence staining of the bone-tissue sections with the adipocyte marker perilipin showed that approximately 85.87% $\pm$ 6.63% of the total tdTom⁺ cells were perilipin⁺ adipocytes after 2 weeks of MPS treatment (Figures 1F and 1G), indicating that most bone-marrow senescent cells were adipocytes during the early stage of glucocorticoid treatment. The percentage of adipocytes out of total senescent cells decreased to approximately 27.31% $\pm$ 11.92% after 4 weeks of MPS treatment (Figures 1F and 1G). Therefore, many other cell types, in addition to adipocytes, became senescent at a later stage of glucocorticoid treatment. Of note, the percentage of perilipin⁺ cells that became senescent (tdTom⁺) was approximately 23.87% $\pm$ 8.29% and 24.53% $\pm$ 6.66% after 2 and 4 weeks of MPS treatment, respectively (Figure 1H), indicating that only a portion of BMAds undergo cellular senescence

(A and B) Representative images of tdTom⁺ cells from proximal and distal femur (A) and analysis of cell number per mm² tissue area (N. tdTom⁺ cells) (B).

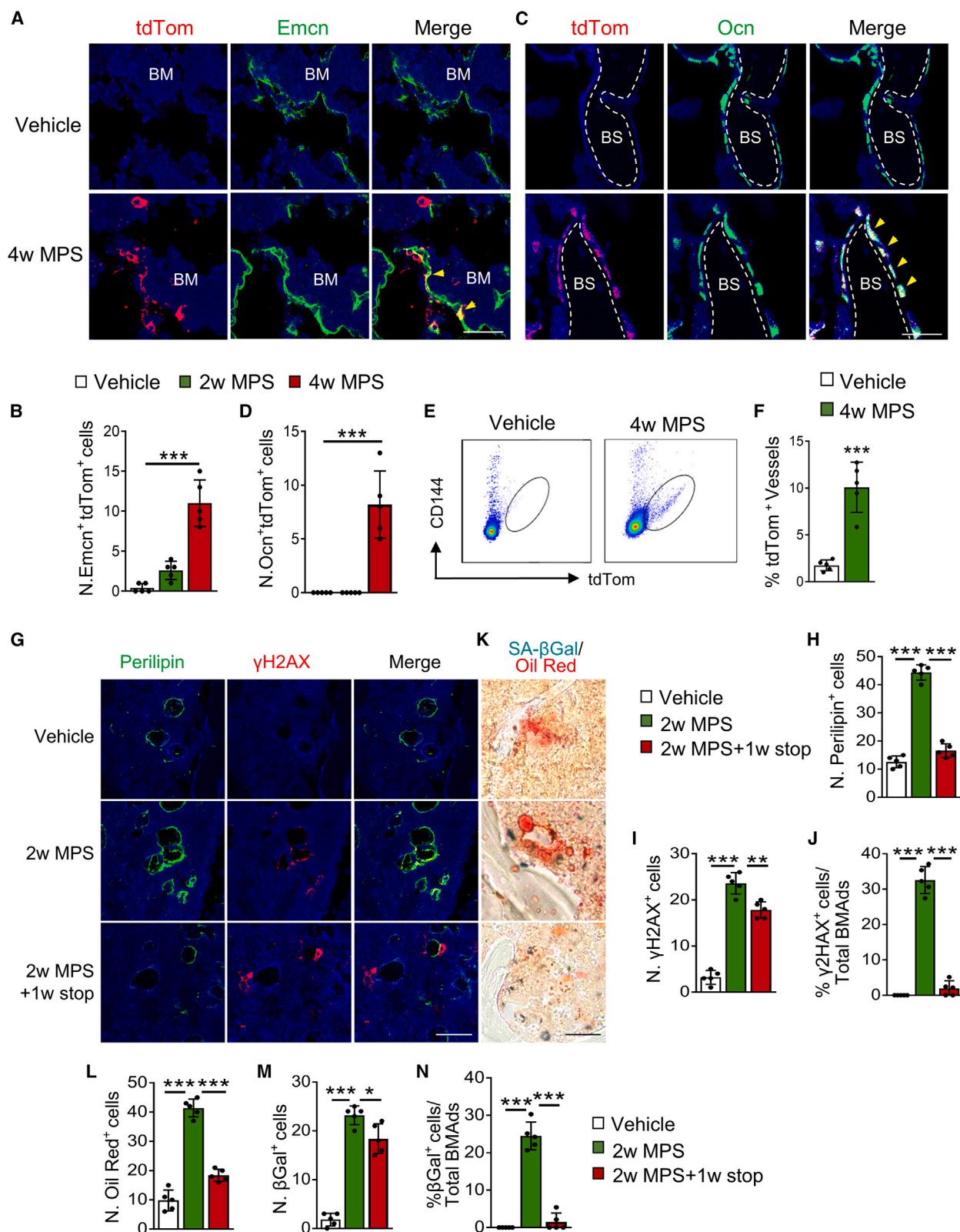
(C) Schematic diagram illustrating the experimental procedure. Femoral bone and bone-marrow tissue was collected from mice and digested, and the isolated cells were subjected to flow cytometry analysis.

(D and E) Representative flow-cytometry images of tdTom-expressing cells of the femoral bone (D) and the percentages of tdTom⁺ cells out of total bone/bone marrow cells (E).

(F–H) Immunofluorescence staining of perilipin in femoral bone sections. Representative images are shown in (F). Quantification of the percentage perilipin⁺ cells in total tdTom-positive cell population (% perilipin⁺ cells/total tdTom⁺ cells) and the percentage of tdTom⁺ cells in total perilipin-positive cell population (% tdTom⁺ cells/total perilipin⁺ cells) are shown in (G) and (H), respectively.

(I and J) 3-month-old C57BL/6 mice were treated with MPS or vehicle for 2 weeks. (I) Fluorescent *in situ* hybridization followed by immunofluorescence staining was performed to identify SADS-positive senescent cells (in red) and perilipin⁺ adipocytes (in green), respectively. (J) Double-immunofluorescence staining of γ H2AX and perilipin in femoral bone sections.

(K–M) 3-month-old C57BL/6J mice were treated with MPS at 10 mg/m²/day or vehicle for different time periods as indicated. Double-immunofluorescence staining of γ H2AX and perilipin in femoral bone sections. Representative images are shown in (K). Quantification of the numbers of perilipin-positive cells per mm² tissue area (N. Perilipin⁺ cells) and the number of γ H2AX-positive cells per mm² tissue area (N. γ H2AX⁺ cells) are shown in (L) and (M), respectively. (n = 5 mice.) Data are represented as mean $\pm$ S.E.M. *p < 0.05, **p < 0.01, ***p < 0.001, as determined by two-tailed Student's t tests for 2 groups and one-way ANOVA for 3 or more groups.



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in response to MPS treatment. We also detected the SADSs and the DNA damage marker γ H2AX, robust markers of senescent cells,^{48–51} and found that SADS⁺ and γ H2AX⁺ signals were localized primarily in perilipin⁺ BMAds of MPS-treated mice (Figures 1I and 1J). We also examined whether shorter periods of glucocorticoid treatment could induce cellular senescence. The number of perilipin⁺ BMAds started rising at 1 week after MPS treatment, whereas γ H2AX⁺ senescent cells were higher only at 2 weeks but not at the earlier time points after MPS treatment relative to the vehicle treatment group (Figures 1K–1M). Therefore, senescence was not induced before the increase of bone-marrow adiposity, further confirming that most of the senescent cells are BMAds. Similarly, co-staining of bone-tissue sections with SA- β Gal and oil red O, a marker of mature adipocytes, showed markedly greater SA- β Gal/oil red O-double positive cells in MPS-treated mice relative to vehicle-treated mice (Figures S1G and S1H). Therefore, adipocytes are the first cell type to show a senescent phenotype in the bone/bone-marrow microenvironment following glucocorticoid treatment.

Senescent vascular cells and OBs accumulate in mouse bone and BM in a late phase of glucocorticoid treatment

Because we observed many more senescent cells after 4 weeks (vs. 2 weeks) of MPS treatment, we attempted to identify the major cell types that undergo cellular senescence during this later phase of MPS treatment. We previously found that bone vascular-endothelial cells are a vulnerable cell type to glucocorticoid-induced cellular senescence.²⁷ We examined whether 4-week MPS treatment also induces the accumulation of senescent vascular-endothelial cells in adult mice. As expected, we found a much greater number of endomucin (Emcn)⁺ vascular-endothelial cells that express tdTom in the distal femur of $p16^{tdTom}$ mice after 4 weeks of MPS treatment (Figures 2A, 2B, and S2H). We noticed that some tdTom⁺ cells were OB-like and were present on the bone surface (BS) in a closely packed layer. Indeed, the number of tdTom-expressing osteocalcin (OCN)⁺ OBs were markedly greater in the distal femur of mice with 4-week MPS treatment vs. vehicle treatment (Figures 2C and 2D). Notably, almost none of the tdTom-expressing cells were Emcn⁺ vascular-endothelial cells or OCN⁺ OBs after 2 weeks of MPS treatment (Figures 2A–2D). These results suggest that the accumulation of senescent vascular cells and OBs is a later response to MPS treatment relative to the senescence of BMAds.

Further, flow-cytometric analysis of the femoral bone and bone-marrow cells of $p16^{tdTom}$ mice also showed a dramatically higher incidence of tdTom⁺ cells in the CD144⁺ cell population, which includes vascular-endothelial cells and endothelial progenitor cells, in the 4-week MPS-treated group vs. the vehicle-treated group (Figures 2E and 2F). Consistent with these findings, the total number of SA- β Gal⁺ cells and the percentage of SA- β Gal-expressing endothelial cells were also much higher in the 4-week MPS-treated mice vs. vehicle-treated mice (Figures S2A–S2C). We sorted the CD144⁺ bone vascular-endothelial cells using fluorescence-activated cell sorting (FACS) followed by quantitative real-time polymerase chain reaction analysis. CD144⁺ cells from MPS-treated mice (vs. vehicle-treated mice) had reduced expression of the cell proliferative marker *Mki67* (Figure S2D) and higher expressions of *Cdkn2a* (Figure S2E) and *Tp53* (Figure S2F), but not *Cdkn1a* (Figure S2G). Therefore, bone vascular cells and OBs may undergo secondary senescence after the primary BMAd senescence induced by glucocorticoid treatment. To further validate this assumption, we tested whether senescent cells could still be detected in the bone tissue of mice after stopping glucocorticoid treatment. The number of perilipin⁺ BMAds was greater in 2-week MPS-treated mice (vs. vehicle group) when the mice were sacrificed immediately after treatment. However, the number of perilipin⁺ BMAds were similar to the level of the vehicle-treated control group when the mice were sacrificed a week after the termination of treatment (Figures 2G and 2H), indicating a regression of BMAd accumulation. Oil red O staining of the bone tissue sections showed a consistent result (Figures 2K and 2L). Importantly, the numbers of γ H2AX⁺ (Figure 2I) and SA- β Gal⁺ (Figure 2M) senescent cells remained higher in this group compared with the vehicle group. Most of the senescent cells were non-BMAds because the senescence markers γ H2AX (Figures 2G and 2J) and SA- β Gal (Figures 2K and 2N) were not overlapping with BMAd markers. Therefore, although both the expansion of bone-marrow adipose tissue and the senescent BMAds regressed after termination of MPS treatment, senescence likely spread to other cell types through BMAd-produced SASP factors.

Glucocorticoids directly activate a cellular senescence program in BMAd lineage cells

We then tested whether glucocorticoid treatment directly induces cellular senescence of BMAd lineage cells. Bone-marrow

Figure 2. Senescent vascular cells and osteoblasts accumulate in murine bone and bone marrow in a late phase of glucocorticoid treatment

3-month-old $p16^{tdTom}$ mice were treated with MPS at 10 mg/m²/day or vehicle for different time periods as indicated.

(A–D) Immunofluorescence staining of Endomucin (Emcn) (A) or osteocalcin (Ocn) (C) in femoral bone sections. Yellow arrow heads: cells are tdTom⁺ and Emcn-double positive in (A) or tdTom⁺ and Ocn-double positive in (C). BM, bone marrow; BS, bone surface. Quantified numbers of Emcn- and tdTom-double positive cells per mm² tissue area (N. Emcn⁺tdTom⁺ cells) and Ocn- and tdTom-double positive cells per mm² tissue area (N. Ocn⁺tdTom⁺ cells) are shown in (B) and (D), respectively.

(E and F) Bone and bone marrow cells were isolated from femoral bone for flow-cytometry analysis. Representative images are shown in (E). Percentages of tdTom-expressing cells in total CD144⁺ vascular cell population are shown in (F). 3-month-old C57BL/6J mice were treated with MPS at 10 mg/m²/day or vehicle for different time periods as indicated.

(G–J) Double-immunofluorescence staining of γ H2AX and perilipin in femoral bone sections. Representative images are shown in (G). Quantification of the number of perilipin-positive cells per mm² tissue area (N. Perilipin⁺ cells), the number of γ H2AX-positive cells per mm² tissue area (N. γ H2AX⁺ cells), and the percentage of γ H2AX⁺ cells in total perilipin-positive cell population (% γ H2AX⁺ cells/total BMAds) in (H)–(J), respectively.

(K–N) Representative images of SA- β Gal and oil red staining of femoral bone sections are shown in (K). Quantification of the numbers of oil red⁺ cells per mm² tissue area, SA- β Gal⁺ cells per mm² tissue area, and the percentage of SA- β Gal⁺ cells in total oil red⁺ BMAds are shown in (L)–(N), respectively. (n = 5 mice.) Data are represented as mean $\pm$ SEM. *p < 0.05, **p < 0.01, ***p < 0.001 as determined by two-tailed Student's t tests for 2 groups and one-way ANOVA for 3 groups.

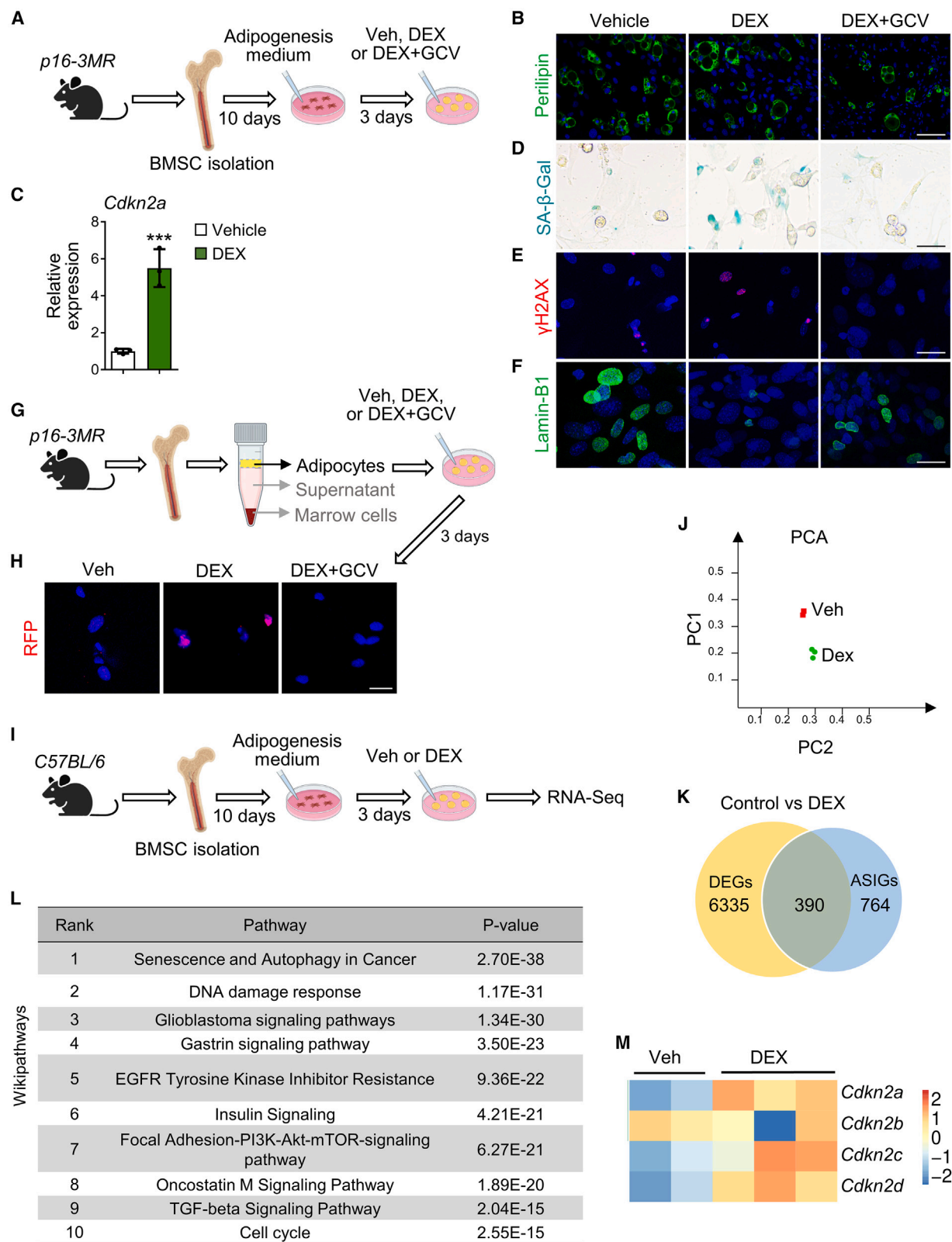


Figure 3. Glucocorticoids directly activate a cellular-senescence program in cultured BMAd lineage cells

(A–F) Bone-marrow mesenchymal stromal cells (BMSCs) were isolated from 3-month-old *p16-3MR* mice and differentiated into bone marrow adipocytes/preadipocytes with adipogenesis medium for 10 days (A). The differentiated adipocytes and preadipocytes were then stimulated with vehicle, dexamethasone

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mesenchymal stromal cells (BMSCs) were isolated from mice and cultured in various conditions where the cells underwent different stages of adipogenic differentiation. We first evaluated if glucocorticoid treatment induces senescence of BMSCs and early-stage adipocyte precursors during adipogenic differentiation. To do this, we cultured BMSCs in a growth medium, in which BMSCs maintain their undifferentiated state, and a synthetic glucocorticoid dexamethasone (DEX) was added to the medium for different time periods (Figure S3A). mRNA expressions of key adipogenesis marker genes, including *Pparg*, *Cebpa*, and *Adipoq*, were elevated rapidly (i.e., 1 or 3 days) after DEX treatment relative to vehicle-treatment controls (Figure S3C). However, perilipin⁺ mature adipocytes were almost undetectable at all time points after DEX treatment (Figure S3B). Thus, although the cells were committed/differentiated to the adipocyte lineage, most of them were premature adipocyte precursors even at 7 days after DEX treatment. We then tested if the expression of cellular senescence markers increased in the cells. Increased *Cdkn2a* (p16INK4A encoding gene) expression (Figure S3D) and SA-βGal⁺ cells (Figure S3E) were detected only at 7 days, but not at 1 and 3 days, after DEX treatment vs. the vehicle control group. Moreover, cells that were positive for γH2AX and that lost lamin-B1 expression—hallmark changes for cells that have progressed into full senescence^{52,53}—were not detected in all three treatment groups (Figures S3F and S3G). The results suggest that the cellular senescence program is initiated in the premature adipocyte precursors, but the full senescence phenotype only developed in mature BMAds. Our data also suggests that, in response to glucocorticoid treatment, the senescence program was not induced in uncommitted BMSCs. Supporting the conclusions, flow-cytometry analysis showed that the percentage of p16^{tdTom}⁺ cells was elevated in bone-marrow CD45[−]CD31[−]CD24[−]Sca1⁺ adipocyte precursors (Figures S3H and S3I) but not in the CD45[−]CD31[−]CD24⁺Sca1⁺ BMSC population^{54,55} in *p16^{tdTom}* mice after 2 weeks of MPS treatment (Figures S3J and S3K).

We then examined cellular senescence in the later stages of adipogenesis when most of the cells become mature adipocytes. BMSCs isolated from *p16-3MR* mice, in which the senescent cells can be eliminated by treatment with ganciclovir (GCV),⁵⁶ were first incubated with adipogenesis medium to induce their differentiation into preadipocytes and adipocytes. The cells were then treated with DEX with or without GCV (Figure 3A). Most of the cells became mature adipocytes as approximately 56.43% ± 3.52% of the cells were perilipin⁺ (Figure 3B). DEX-treated cells had dramatically increased *Cdkn2a* expression (Figure 3C) and SA-βGal expression (Figure 3D) relative to vehicle-treated cells. Importantly, in contrast to pre-

mature adipocyte precursors (Figures S3F and S3G), adipocytes at the later stages of adipogenesis exhibited full senescence phenotype, including γH2AX expression and loss of lamin-B1 (Figures 3E and 3F). Of note, DEX-induced SA-βGal⁺ cell accumulation, γH2AX expression, and loss of lamin-B1 were all abolished by co-treatment with GCV (Figures 3D–3F), suggesting that they are senescence-dependent features. We also isolated BMAds directly from *p16-3MR* mice with an established centrifuge-based method and subsequently treated the adipocytes with DEX with or without GCV (Figure 3G).⁵⁶ Increased p16-RFP⁺ cells, which represent senescent cells in *p16-3MR* mice, were detected only in the DEX treatment group and not in the vehicle group or DEX and GCV co-treatment group (Figure 3H), confirming the glucocorticoid-induced senescence of BMAds. We also tested whether glucocorticoids can directly induce OB and endothelial-cell senescence *in vitro* by treating calvarium-originated OBs or human umbilical-vein endothelial cells (HUVECs) with DEX. Surprisingly, DEX failed to increase SA-βgal expression of these two cell types (Figures S3L and S3M). These results suggest that the BMAd lineage, but not OBs and vascular-endothelial cells, are the direct target of glucocorticoid-induced cellular senescence.

To further confirm that glucocorticoids directly induce the senescent phenotype of BMAd lineage cells and to determine the underlying molecular mechanisms, we conducted RNA-seq analysis of the *in vitro* differentiated adipocytes and preadipocytes (Figure 3I). Principal component analyses (PCAs) revealed that mRNA expression profiles can effectively differentiate DEX-treated cells from the vehicle-treated controls (Figure 3J). After the screening of differentially expressed mRNAs by the filter criteria ($p < 0.05$), 6,725 differentially expressed genes (DEGs) were identified in the DEX- vs. vehicle-treated cells (Table S1). Comparison of our data with previously defined aging/senescence-induced genes (ASIGs)⁵⁷ identified a total of 390 ASIGs in the DEX- vs. vehicle-treated cells (Figure 3K; Table S1). WikiPathway analysis showed that significantly differentially expressed ASIGs have influenced “senescence and autophagy in cancer,” “DNA damage repair,” “cell cycle,” and “TGF-β signaling pathway,” which were notable for their known links to cellular senescence (Figure 3L). The most significantly upregulated ASIGs in DEGs are listed in Figure S4. Of note, the key senescence genes *Cdkn2a*, *Cdkn2c*, and *Cdkn2d*, which are p16INK4a, p18INK4c, and p19INK4d encoding genes, respectively, are among the significantly upregulated genes (Figure 3M), supporting the notion that INK-family proteins are major mediators of glucocorticoid-induced BMAd senescence.

(DEX), or DEX + ganciclovir (GCV) for 3 days. Cells were fixed and subjected to perilipin immunostaining (B), *Cdkn2a* (p16 encoding gene) mRNA measurement by quantitative real-time polymerase chain reaction (C), SA-βGal staining (D), γH2AX immunostaining (E), and Lamin-B1 immunostaining (F), respectively. (G and H) BMAds were isolated from 3-month-old *p16-3MR* mice as shown in (G) and were treated with vehicle, DEX, or DEX + GCV for 3 days. The cells were then subjected to RFP staining with representative pictures in (H).

(I–M) BMSCs were isolated from C57BL/6 mice and differentiated into bone-marrow adipocytes and preadipocytes with adipogenesis medium for 10 days (I). The differentiated adipocytes/preadipocytes were stimulated with vehicle or DEX and subjected for bulk RNA-seq. Principal-component analysis (PCA) of the DEG expression profile of sequenced adipocyte samples is shown in (J). Venn map analyzing the numbers of DEGs overlap with previously identified aging and senescence ASIGs (K). Top enriched pathways of ASIGs genes significantly upregulated or downregulated in DEX-treated cells relative to vehicle-treated cells based on RNA-seq data (L). Heatmap showing increased expression of INK-family genes in DEX-treated cells relative to vehicle-treated cells (M).

In vitro experiments were repeated 3 times. Data are represented as mean ± SEM. *** $p < 0.001$ as determined by two-tailed Student's *t* tests.

Glucocorticoids induce senescence of BMA_d lineage cells by activating a prostaglandins-PPAR γ -INK positive feedback loop

BMA_ds are a source of lipids, and they have an active lipid metabolism that regulates bone remodeling and, potentially, whole-body metabolism.^{5,58,59} A recent study revealed an increase in the synthesis of prostaglandin D₂-related oxylipins, such as dihom-15d-PGJ₂, as a key mechanism promoting cellular senescence and the corresponding SASP in fibroblasts.⁴⁵ Because glucocorticoids modulate lipid metabolism in adipocytes,^{22,31,60} we explored whether glucocorticoid treatment is also associated with prostaglandin generation. By analysis of the RNA-seq data, we found that *Ptgds* and *Ptgis*, which encode for prostaglandin D₂ synthase and prostaglandin I₂ synthase, respectively, were significantly upregulated in isolated BMA_ds (Figure 4A). The other two synthases in the family, *Ptges* and *Tbxas1*, which encode for prostaglandin E₂ synthase and thromboxane A synthase, respectively, were downregulated.

These results suggest that the production of PGD₂ and PGI₂ in BMA_ds may increase in response to glucocorticoid treatment. Thus, we used ELISA to measure the concentrations of prostaglandins, including PGJ₂ (a sum of 15d-PGJ₂ and dihom-15d-PGJ₂), PGD₂ (a precursor of 15d-PGJ₂), and PGE₂ in the conditioned medium (CM) in different treatment groups (Figure 4B). As we expected, DEX-treated cells showed elevated PGJ₂ and PGD₂ secretion but unchanged PGE₂ secretion (Figure 4C). It has been reported that 15d-PGJ₂ induces fibroblast senescence.⁴⁵ Consistent with this finding, we found that 15d-PGJ₂ exerted similar effects as DEX on adipocytes and preadipocytes in inducing upregulation of senescence-related genes, including *Cdkn2a* and *Cdkn1b* (p16INK4a and p27KIP1 encoding genes, respectively) (Figure 4D).⁴⁵ *Cdkn1a* (a p21CIP1 encoding gene) levels were not changed by DEX or 15d-PGJ₂ treatment, which is consistent with the findings from Figure S3G.

We also examined whether a reduction in prostaglandin production occurred in response to inhibition of cyclooxygenase 2 (COX2), a key enzyme involved in prostaglandin production, and thus whether such inhibition could antagonize adipocyte and preadipocyte senescence.^{61–63} Whereas the adipocytes and preadipocytes treated with DEX alone had increased SA- β Gal expression, the number of SA- β Gal⁺ cells diminished in cells co-treated with DEX and celecoxib, a selective COX2 inhibitor (Figure 4E). We then measured the levels of key senescence-inducing genes, including *Cdkn2a* and *Cdkn1b*, and *Cdkn1a*, in the cells of different treatment groups.^{34,41} The expressions of *Cdkn2a* and *Cdkn1b*, but not *Cdkn1a*, were elevated in the DEX treatment group and decreased to basal levels in the DEX and celecoxib co-treatment group (Figure 4F). These results suggest that glucocorticoids increase the production of oxylipins, which in turn induce the senescence of BMA_ds. The results from Figures 4D, 4F, and S2G also consistently suggest that p21CIP1 is not involved in BMA_d senescence induced by glucocorticoids.

As 15d-PGJ₂ is a natural agonist of PPAR γ ,^{64–66} we postulated that PPAR γ may be involved in this process. DEX and the PPAR γ agonists 15d-PGJ₂ and rosiglitazone (Rosi) induced *Pparg* gene expression in adipocytes and preadipocytes (Figure 5A). Importantly, co-treatment with celecoxib abolished this effect of DEX (Figure 5B), indicating that increased secretion

of prostaglandins mediates DEX-induced upregulation of *Pparg* expression.

We then tested whether activated PPAR γ leads to cellular senescence. We found that the PPAR γ synthetic agonist Rosi induced dramatic upregulation of senescence marker genes, including *Cdkn2a* and *Cdkn1b*, but not *Cdkn1a* (Figure 5C).^{34,41} Moreover, DEX- and 15d-PGJ₂-induced upregulation of senescence marker genes were largely abrogated by co-treatment of the cells with either of the PPAR γ antagonists T0070907 and GW9662 (Figures 5D and 5E). Consistent with the changes in senescence gene expression, SA- β Gal expression was also greatly stimulated in the adipocytes and preadipocytes treated with PPAR γ agonists 15d-PGJ₂ and TZD compared with vehicle-treated cells (Figure 5F). In contrast, the effects of DEX and 15d-PGJ₂ on inducing SA- β Gal expression were almost abolished by the PPAR γ antagonists T0070907 and GW9662 (Figure 5G). Likewise, RNA-seq data showed that the significantly upregulated INK-family genes induced by DEX were effectively reversed by co-treatment with the PPAR γ antagonist GW9662 (Figure 5H). Intriguingly, DEX-induced upregulation of *Ptgds* and *Ptgis* expression were both blunted by treatment of the cells with the PPAR γ antagonists GW9662 (Figures 5I and 5J). Thus, PPAR γ activation, in turn, promotes PGD₂ and PGI₂ synthesis in BMA_ds. Together, these results suggest that the reciprocal feedback loop of oxylipins and PPAR γ signaling is crucial in initiating and maintaining the senescence program of BMA_ds.

Senescent BMA_ds acquire a SASP that spreads senescence to bone-marrow vascular cells and OBs

We next investigated whether senescent BMA_ds acquire a SASP. We conducted analyses of our RNA-seq data again and found a typical SASP expression profile in the DEX-treated vs. vehicle-treated adipocytes and preadipocytes, as evidenced by upregulation of multiple previously identified SASP factors,⁴³ including inflammatory cytokines, chemokines, growth factors, secreted proteases, and secreted ECM components (Figure 6A). Intriguingly, the effects of DEX on inducing the upregulation of these factors were fully antagonized by the PPAR γ antagonist GW9662, indicating that PPAR γ activation is a key contributor not only to senescence cell cycle arrest but also to SASP development of the BMA_ds.

To substantiate the data from the transcriptomic analysis, we performed proteomics assays. Isolated BMA_ds from *p16-3MR* mice were treated with DEX with or without GCV using the same procedure as illustrated in Figure 3G. Conditioned media from different treatment groups were collected for an antibody array screening analysis to identify the secreted factors in the senescent BMA_ds. In this array, 111 candidate factors, including various types of cytokines and chemokines, were screened. As shown in Figures 6B and 6C, DEX-treated BMA_ds had higher expression of a broad range of the factors tested, whereas the expression of almost all these factors in GCV-co-treated cells became much lower relative to cells treated with DEX, and even reached the levels of the vehicle-treated cells in some cases, suggesting a diminished SASP of BMA_ds by senescent cell clearance. Of note, approximately half of these factors (labeled with red boxes in Figure 6C) were also identified in the RNA-seq analysis, further confirming them as SASP factors.

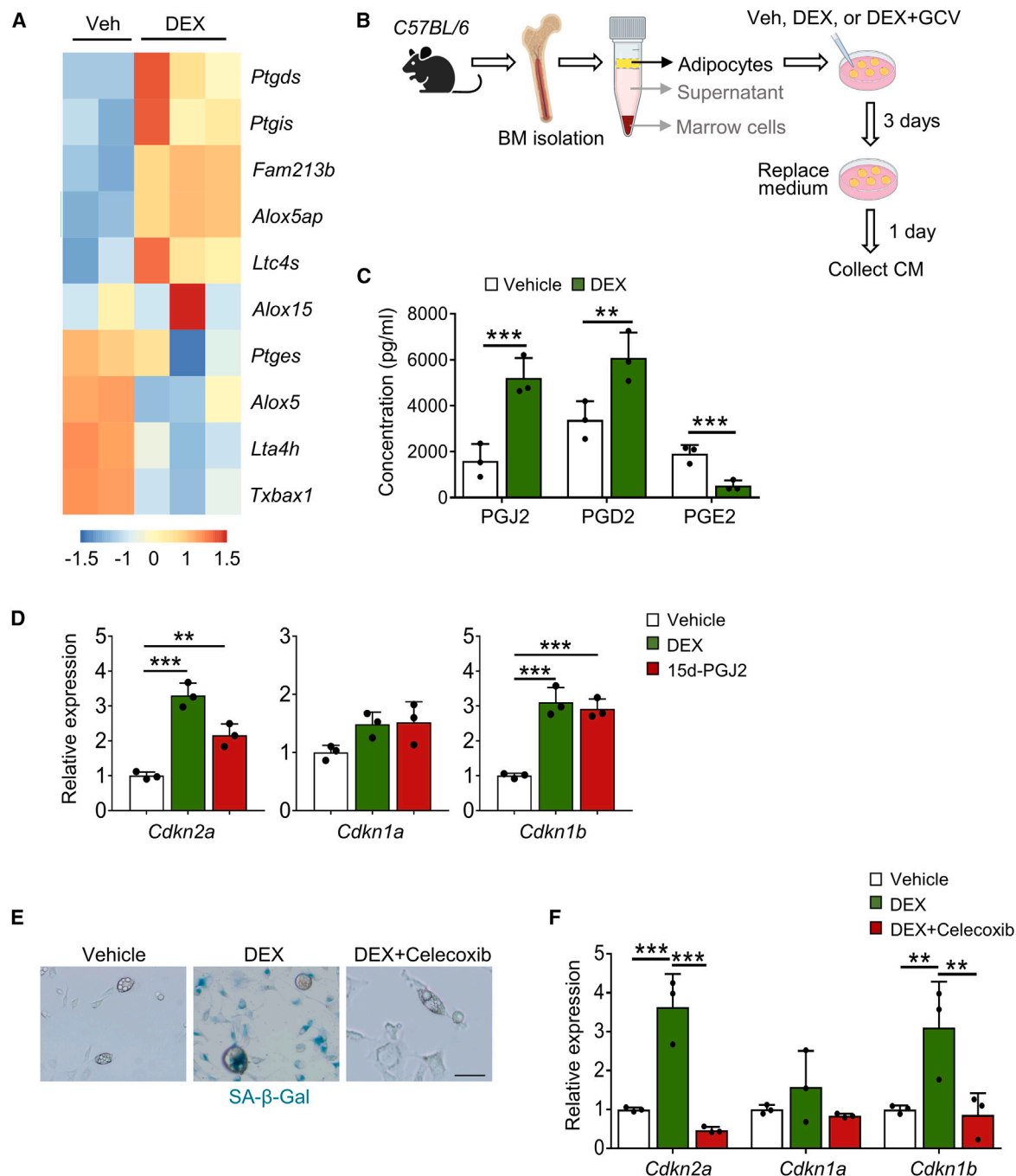


Figure 4. Glucocorticoids induce senescence of BMD lineage cells by activating the prostaglandin pathway

(A) Heatmap of RNA-seq data showing expression changes encoding oxylipin synthesis genes in DEX-treated relative to vehicle-treated adipocytes/pre-adipocytes.

(B) Schematic diagram showing the procedure of the conditioned medium (CM) preparation from cultured BMADs.

(C) The concentrations of prostaglandins (PGJ2, PGD2, and PGE2) in CM were measured by ELISA.

(D) BMADs were isolated and treated with vehicle, DEX, or 15d-PGJ2 for 3 days. mRNA levels of *Cdkn2a*, *Cdkn1a*, and *Cdkn1b* in the cells were measured by quantitative real-time polymerase chain reaction analysis.

(E) Adipocytes and preadipocytes differentiated from BMSCs were treated with vehicle, DEX, or 15d-PGJ2 for 3 days. SA-βGal staining was performed.

(F) Adipocytes and preadipocytes differentiated from BMSCs were treated with vehicle, DEX, or DEX plus celecoxib for 3 days. mRNA levels of *Cdkn2a*, *Cdkn1a*, and *Cdkn1b* in the cells were measured by quantitative real-time polymerase chain reaction analysis.

In vitro experiments were repeated 3 times. Data are represented as mean ± SEM. **p < 0.01, ***p < 0.001 as determined by two-tailed Student's t tests for 2 groups and one-way ANOVA for 3 groups.

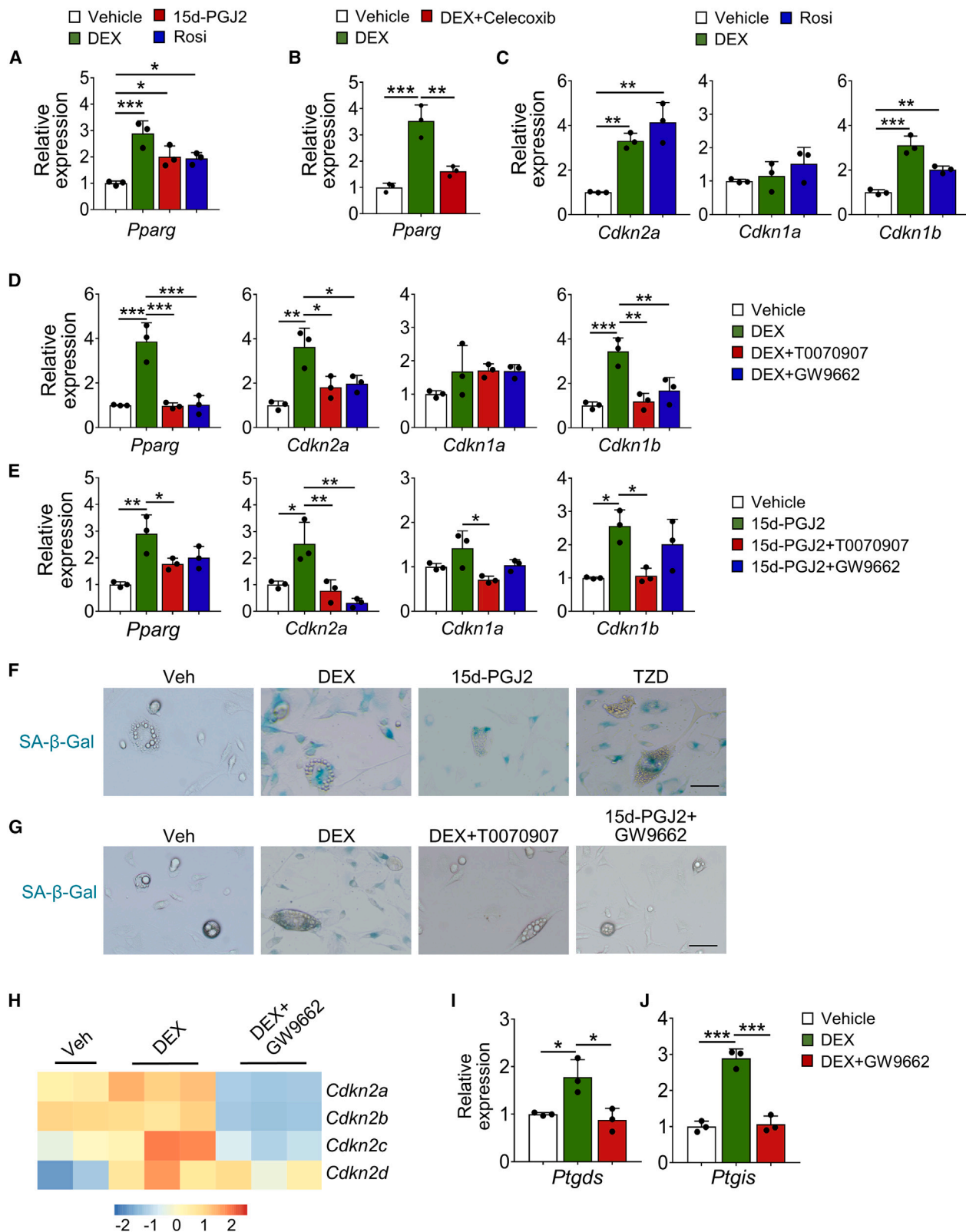


Figure 5. An oxylipins-PPAR γ positive feedback loop mediates glucocorticoid-induced BMAd senescence

(A–E) Adipocytes and preadipocytes differentiated from murine BMSCs were stimulated with the different treatments as indicated. mRNA levels of the indicated genes were measured by quantitative real-time polymerase chain reaction analysis.

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We then tested whether the SASP factors secreted by BMAdS can act on other cell lineages in a paracrine manner. As we detected accumulated senescent vascular-endothelial cells and OBs in mouse bone tissue at a later stage (4 weeks) after MPS treatment *in vivo* (Figure 2), we assessed the effects of BMAd-secreted SASP factors on endothelial cell and OB activities. To do so, we used the combination of dasatinib and quercetin (D+Q), which effectively eliminates senescent cells,⁴⁸ and we prepared conditioned medium (CM) from vehicle-treated BMAdS (Veh CM) and DEX-treated BMAdS in the absence (DEX CM) or presence (DEX+DQ CM) of the senolytic drug D+Q. The conditioned media were added to the cultures of endothelial cells and OBs (Figure 6D). Both endothelial cells and OBs had significantly higher SA-βGal expression when incubated with DEX CM compared with Veh CM. CM prepared from senescent cell-eliminated cells (DEX+DQ CM) failed to induce SA-βGal expression in the endothelial cell and OB cultures (Figure 6E). Western blot analysis showed that DEX CM induced the expression of the senescence marker p16INK4a, whereas conditioned media prepared from BMAdS with DEX and senolytic drug co-treatment (DEX+DQ CM) failed to induce p16INK4a expression in endothelial cells and OBs (Figure 6F). To test whether the senescence-associated BMAd secretome directly causes bone loss and widespread cellular senescence in bone and BM, 3-month-old C57BL/6 mice were treated with vehicle and MPS for 2 weeks (donors), and the mature BMAdS and CD45⁺CD31⁺CD24⁺Sca1⁺ adipocyte precursor cells (APCs) were separately isolated from the mice and transplanted into the femoral bone-marrow cavity of 3-month-old wild-type (WT) mice (acceptors) (Figures 6G and S5A). 4 weeks after transplantation, the bone phenotype and cellular senescence were examined. By micro-CT analyses, we found that the acceptors transplanted with BMAdS from MPS-treated donors (MPS-BMAd), compared with those transplanted with the same number of BMAdS from vehicle-treated donors (Veh-BMAd), exhibited a low-bone-mass phenotype (Figure 6H), reduced bone volume/tissue volume (BV/TV) (Figure 6I) and trabecular thickness (Tb. Th) (Figure 6J), unchanged trabecular numbers (Tb. N) (Figure 6K), and increased trabecular separation (Tb. Sp) (Figure 6L). In contrast, the acceptors transplanted with APCs isolated from MPS-treated donors (MPS-APCs) did not show evident phenotypic bone changes relative to those transplanted with APCs from vehicle-treated donors (Veh-APCs). Histological analyses showed dramatically increased SA-βGal⁺ senescent cells (Figures S5B and S5C) and loss of type-H vessels (Figures S5D and S5E) in bone and BM of the MPS-BMAd (vs. the Veh-BMAd) group. The impairment of bone angiogenesis, as evidenced by the loss of type-H vessels, was not observed in the MPS-APC (vs. the Veh-APC) group although SA-βGal⁺ senescent cell numbers were a little higher in MPS-APC relative to the Veh-APC group. These results suggest that senescent BMAdS, but not the premature

adipocyte precursors, are sufficient to cause secondary senescence in bone and BM and the resultant bone deterioration.

Targeting senescent BMAdS or the SASP attenuates glucocorticoid-induced bone loss

We asked whether inhibiting the initial senescence of BMAdS could prevent glucocorticoid-induced secondary senescence of bone vascular cells and OBs. To achieve this purpose, we generated conditional *Adipoq-Cre::p16^{flox/flox}* (p16-cKO) mice, in which *p16INK4a* is deleted selectively in adipocytes. 2 weeks of MPS treatment in *p16^{flox/flox}* (WT) mice induced senescence of BMAdS, as indicated by SA-βGal-expressing adipocytes that were also oil red O-positive. However, senescent SA-βGal⁺ BMAdS were undetectable in p16-cKO mice treated with MPS for the same duration (Figures 7A and 7B), indicating a blockage of BMAd senescence by *p16INK4a* deletion from the cells. We then evaluated whether the secondary senescence induced by 4 weeks of MPS treatment can be prevented in p16-cKO mice. Indeed, 4-week MPS treatment in WT mice induced increases in the numbers of SA-βGal⁺Emcn⁺ double-positive cells (senescent vascular-endothelial cells) (Figures S6A and S6B). However, the accumulation of these senescent cells was largely abrogated in p16-cKO mice treated with MPS for the same duration. As a result, the numbers of type-H vessels (Figures S6C and S6D) and Ocn⁺ mature OBs (Figures S6E and S6F) were significantly increased in p16-KO mice compared with WT littermates after MPS treatment, suggesting improved angiogenesis and osteogenesis in p16-KO mice. By micro-CT analysis we found that the MPS-induced low-bone-mass phenotype (Figure 7C), characterized by lower BV/TV (Figure 7D) and Tb. Th (Figure 7E), and unchanged Tb. N (Figure 7F), as well as an increase in Tb. Sp (Figure 7G), was partially rescued by p16 knockout in adipocytes. We are aware that *Adipoq-Cre*-driven p16 deletion may also inhibit adipocyte senescence in peripheral tissues. To further test if the SASP from peripheral adipose tissue may also be involved in glucocorticoid-induced bone defects, we transplanted BMAdS isolated from MPS- or vehicle-treated *Adipoq-Cre; p16^{flox/flox}* mice (p16-cKO donors) into the femoral bone-marrow cavity of C57BL/6 mice (acceptors) (Figure 7H). BMAdS from MPS-treated *p16^{flox/flox}* mice (WT donors) caused a bone-loss phenotype in acceptors relative to those transplanted with BMAdS from vehicle-treated WT donors. Intriguingly, BMAdS from MPS-treated p16-cKO donors failed to induce similar low-bone-mass phenotype in the acceptors (Figures 7I–7M). The results also suggest that the peripheral adipose tissue is not a major player in this process.

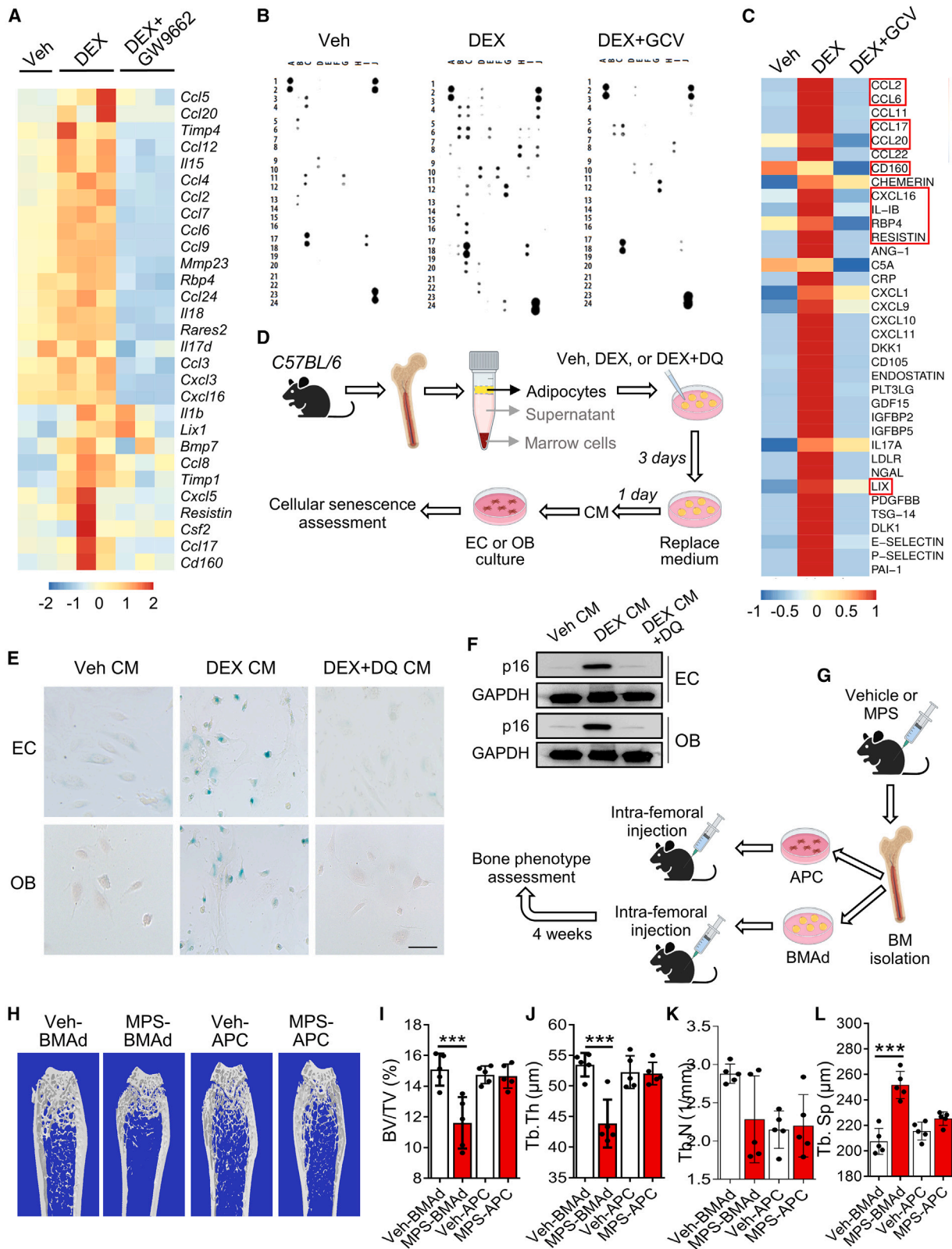
Finally, to further validate the importance of BMAd-produced SASP factors in mediating the adverse effect of glucocorticoids on bone, we tested whether inhibiting the SASP using a pharmacological approach during the early phase of MPS treatment might alleviate glucocorticoid-induced bone deficits. To do so, we used the JAK inhibitor (JAKi) ruxolitinib, which is known to inhibit the SASP of senescent cells.^{53,67} Mice were treated with

(F and G) Adipocytes/preadipocytes differentiated from murine bone marrow MSCs were treated with different reagents as indicated and subjected to SA-βGal staining.

(H) Heatmap of RNA-seq data showing the changes in INK-family genes in cells with the indicated treatments.

(I and J) Quantitative real-time polymerase chain reaction analysis of *Ptgds* and *Ptgis* levels in BMAdS with the indicated treatments.

Experiments were repeated 3 times. Data are represented as mean ± SEM. *p < 0.05, **p < 0.01, ***p < 0.001, as determined by one-way ANOVA.



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vehicle, MPS, or MPS with JAKi, as illustrated in Figure S6G. Representative micro-CT images of distal femur microarchitecture showed gross observations of bone loss in the mice treated with MPS alone for 4 weeks, but much improved bone mass in those co-treated with JAKi for the first 2 weeks (Figure S6H). Quantification of the parameters in the trabecular bone of the distal femur showed significantly reduced BV/TV, Tb. Th, and Tb.N, but increased Tb. Sp, in the MPS group relative to the vehicle-treated group. These parameters were largely rectified by JAKi co-treatment (Figures S6I–S6L). Together, the *in vivo* data provide evidence to support the role of senescent BMAd in inducing bone deterioration through a paracrine effect.

DISCUSSION

It is now widely accepted that primary senescent cells induced by direct insults to the cells can induce secondary senescence in neighboring cells through SASP.^{68,69} Our study shows that glucocorticoid treatment induces primary senescence in a small number of BMAd lineage cells, which spread senescence to other bone and bone-marrow cells by secreting SASP factors, leading to an accumulation of senescent cells and impaired bone microenvironment (Figure 7N). Moreover, we have identified a positive interacting feedback loop of 15d-PGJ2-PPAR γ -INK signaling that triggers and maintains the primary senescence of the BMAd lineage cells. We also identified a panel of SASP factors from BMAd that induce the secondary senescence of bone vascular-endothelial cells and OBs. Inhibiting the primary BMAd senescence or suppressing the SASP alleviated glucocorticoid-induced bone loss. Hence, the senescent BMAd lineage cells may be a new biological target to treat glucocorticoid-associated bone disorders, such as osteoporosis and osteonecrosis.

Our data demonstrate that glucocorticoids induce primary senescence of BMAd lineage cells, characterized by multiple cellular senescence markers, including p16INK4a, SA- β Gal, DNA damage marker γ H2AX, and SADS—as well as loss of nuclear expression of HMGB1 and nuclear envelope protein lamin-B1. More importantly, out of the DEGs in the glucocorticoid- vs. vehicle-treated adipocytes and preadipocytes, 390 genes are aging- and cellular-senescence-associated, strongly supporting the activation of a cellular-senescence program in glucocorticoid-treated BMAd lineage cells. Of note, the INK-family members *Cdkn2a*, *Cdkn2c*, and *Cdkn2d*, which encode for p16INK4a, p18INK4c, and p19INK4d, respectively, but not *Cdkn1a*, which encodes for p21CIP1, were upregulated in glucocorticoid-treated BMAd lineage cells. It was recently identified that p16^{high} senescent cells and p21^{high} senescent cells in fat tissue are two distinct cell populations in the context of obesity.⁷⁰

Our results suggest that INK-family proteins, but not p21, are likely BMAd senescence-inducing factors that are upregulated in response to glucocorticoid treatment.

Our results showed that glucocorticoid-induced BMAd senescence is initiated in the committed adipocyte precursors, as evidenced by a gradual increase in p16INK4a expression and the appearance of SA- β Gal during the adipogenesis of BMSCs. However, our findings from the *in vitro* adipogenic differentiation culture and *in vivo* transplantation experiments strongly suggest that mature BMAd, but not adipocyte precursors, display the features of full senescence,^{52,53} such as dramatically high p16INK4a expression, DNA damage marker γ H2AX expression, nuclear structural change represented by lamin-B1 loss, and the paracrine effects via SASP. Cellular senescence is a highly dynamic, multistep process.^{52,53} It is generally accepted that the initiating step is the transition of temporal to stable cell-cycle arrest through activation of the p53-p21 and/or p16INK4a pathways. The progression to full senescence is characterized by lamin-B1 reduction-associated nuclear deformation, chromatin remodeling, and the secretory phenotype (i.e., SASP). Therefore, glucocorticoids activate a progressive cellular senescence program in BMAd lineage cells, with mature BMAd having a full senescence phenotype. Of note, the elevation of adipogenesis marker genes, including *Pparg*, *Cebpa*, and *Adipoq*, occurred earlier than the appearance of the cellular senescence marker *Cdkn2a* in the BMSC culture assays. Moreover, glucocorticoid treatment did not induce increased expression of senescence markers in uncommitted BMSCs. Therefore, the senescence program is activated and completed much later during the process of BMSC adipogenesis.

We identified a molecular mechanism underlying glucocorticoid-induced senescence of BMAd lineage cells. MPS treatment induces elevated production of oxylipins, particularly PGD2 and its derivative 15d-PGJ2, in BMAd. Although we had technical difficulties in detecting intracellular dihydro-15d-PGJ2 levels, our findings that 15d-PGJ2 is sufficient to induce BMAd senescence and that a selective COX2 inhibitor antagonized BMAd senescence induced by DEX agrees with a recent report that PGD2-related oxylipins, especially dihydro-15d-PGJ2, promote cellular senescence and the SASP.⁴⁵ We further found that these oxylipins activate PPAR γ , which is both necessary and sufficient to induce BMAd senescence. It has been reported that PPAR γ accelerates cellular senescence by inducing *p16INK4a* expression in human diploid fibroblasts.⁷¹ Consistent with this finding, we found that both the PPAR γ endogenous agonist 15d-PGJ2 and PPAR γ synthetic agonists induce dramatic upregulation of the key senescence-inducing factors *p16INK4a* and *p27KIP1*, whereas the knockdown of PPAR γ in BMAd or PPAR γ

Figure 6. Senescent BMAd acquire a SASP that spreads senescence to bone marrow vascular cells and osteoblasts

(A) Heatmap of RNA-seq data showing SASP gene expression in adipocytes/preadipocytes treated with vehicle, DEX, or DEX+GW9662. (B and C) BMAd were isolated from 3-month-old *p16-3MR* mice, cultured in α -minimal essential medium (α MEM) for 1 day, and stimulated with vehicle, DEX, or DEX+GCV for 3 days. The secretion of cytokines in the CMs were measured (B). Heatmap of differentially expressed cytokines in CM is shown in (C). (D) A schematic diagram illustrating the procedure of the CM-based senescence assays. (E and F) Calvarium osteoblasts (OBs) or HUVECs (EC) were cultured with CM prepared from (D). SA- β Gal staining of the cells (E). Western blot analysis of p16 protein expression (F). (G) Schematic diagram showing the experimental design for transplantation. (H–L) Representative micro-CT images in (H) and quantitative analysis of trabecular bone volume/tissue volume (BV/TV) (I), trabecular thickness (Tb. Th) (J), trabecular number (Tb. N) (K), and trabecular separation (Tb. Sp) (L). (n = 5 mice.) Data are represented as mean $\pm$ SEM. ***p < 0.001, as determined by two-tailed Student's t tests for 2 groups.

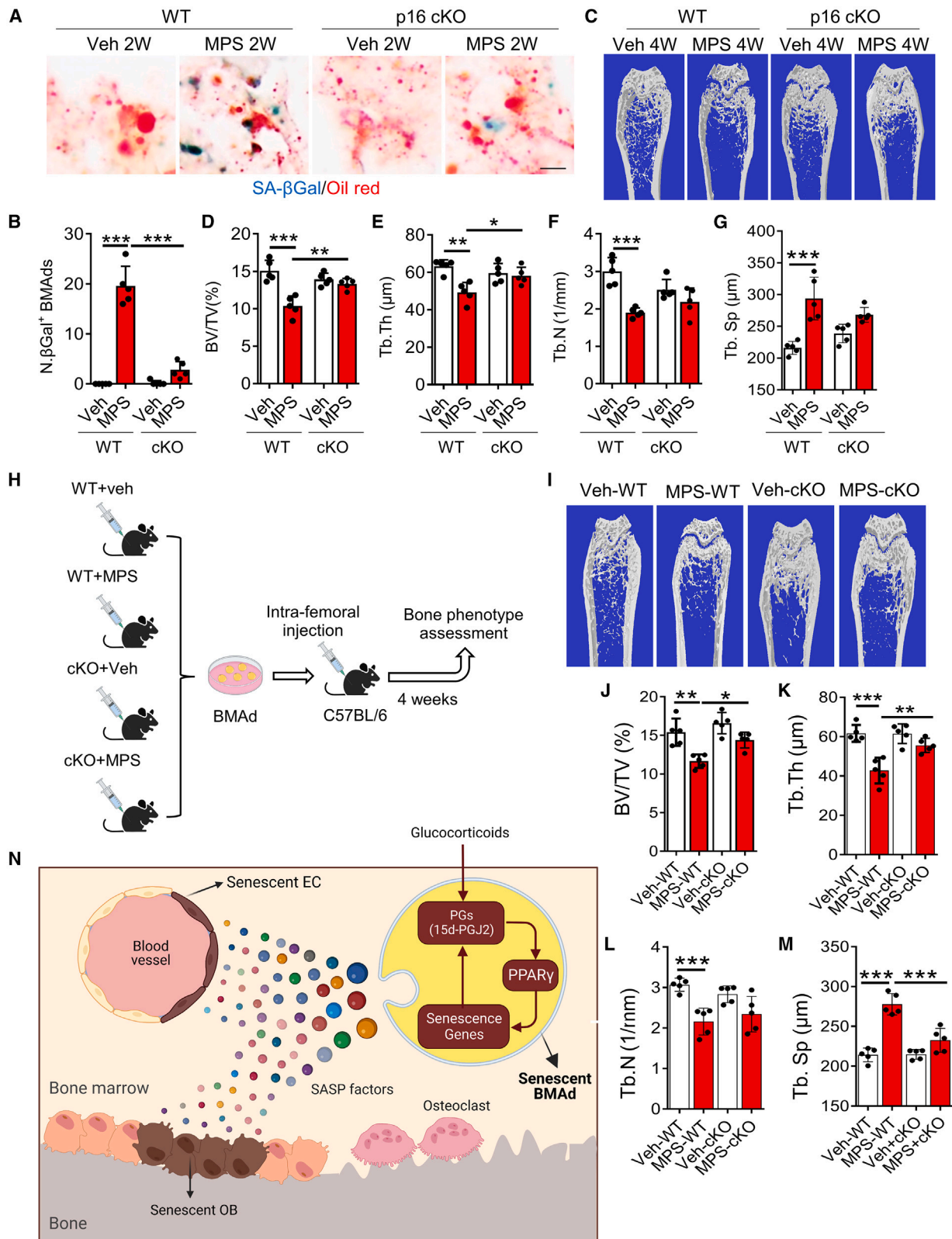


Figure 7. Targeting senescent BMAds rescues glucocorticoid-induced bone loss

(A–G) 3-month-old *AdipoQ-Cre::p16 flox/flox* (p16 cKO) mice and *p16 flox/flox* (WT) mice were injected with daily MPS treatment at 10 mg/m²/day or vehicle for 2 weeks. Representative images of SA-βGal staining and oil red staining in femoral bone tissue sections (A). Quantification of the number of SA-βGal-expressing

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antagonism inhibits the upregulation of these genes induced by DEX treatment. In fact, our RNA-seq result shows that the PPAR γ antagonist GW9662 normalizes the expression levels of many more cellular-senescence-associated genes that were up-regulated in DEX-treated cells, further confirming the key role of PPAR γ activation in mediating BMA δ senescence. Moreover, we provide evidence that PPAR γ activation is also necessary and sufficient to upregulate the gene expression of *Ptgds* and *Ptgis* in BMA δ s. Therefore, oxylipins and PPAR γ form a strong positive feedback loop, contributing substantially to cellular senescence of BMA δ s. This finding suggests that even subtle alterations in this signaling circuit can be amplified, resulting in rapid cellular senescence.

Our *in vitro* and *in vivo* data suggest that glucocorticoid treatment directly activate a senescence-associated program in BMA δ lineage cells but not in other cell types, such as un-committed BMSCs, vascular-endothelial cells, and OBs. Moreover, our data suggest that the secondary senescence of bone-vascular cells and OBs is indirectly mediated through SASP factors secreted by the senescent BMA δ s. Our RNA-seq and protein array studies consistently reveal broadly increased levels of growth factors, cytokines, and chemokines in DEX-treated BMA δ s, all of which were downregulated after clearance of the senescent cells. This finding suggests that the SASP of BMA δ s is an important contributor to the paracrine function of BMA δ s under pathological conditions. Indeed, our *in vitro* results show that CM prepared from the senescent BMA δ s induced the senescence of vascular-endothelial cells and OBs, whereas the senolytic drugs D+Q blunted the senescence of both cell types, validating the role of the SASP of BMA δ s in inducing the secondary senescence of the neighboring cells.

Our *in vivo* results further demonstrate that short-term administration (i.e., 2 weeks) of the SASP inhibitor (JAKi) also largely rescues the low-bone-mass phenotype in mice treated with longer-term glucocorticoid (4 weeks). Moreover, blockade of primary BMA δ senescence by deleting *p16* selectively in BMA δ s almost fully abolished secondary senescence of the bone vascular-endothelial cells and OBs in glucocorticoid-treated mice. We are aware that the rescued bone phenotype may not be solely attributed to the inhibition of BMA δ senescence as AdipoQ-Cre has been used to label not only mature adipocytes but also adipocyte precursors and multipotent BMSCs.^{72–74} However, our *in vitro* and *in vivo* experiments showed that senescence was not induced in BMSCs in response to glucocorticoid treatment. More importantly, our transplantation experiments clearly showed that transplanting mature adipocytes—but not APCs isolated from MPS-treated mice—results in secondary senescence in BM and a bone-loss phenotype. The results suggest that mature BMA δ s with full senescence features are

sufficient to spread the senescence in bone and the bone-marrow microenvironment for bone loss, whereas immature adipocyte precursors do not have this capacity.

It appears that the primary bone and bone-marrow cell types undergoing cellular senescence in response to glucocorticoid treatment are age and location dependent. We recently revealed that in growing young mice, glucocorticoid treatment induces vascular-endothelial cell senescence in the metaphysis of long bone by reducing the number of osteoclasts and the secreted ANG.²⁷ During childhood, the metaphysis of long bone has abundant osteogenesis-coupled type-H vessels with actively proliferative endothelial cells, whereas type-H vessels diminish gradually in adulthood.⁷⁵ Conversely, there is much less bone-marrow adipose tissue at young ages, as it accumulates with aging.⁵⁴ It is not surprising that vascular cells in the metaphysis of growing bone are the primary cell type that becomes senescent, whereas BMA δ s undergo more repaid senescence than other cell types in adult bone in response to glucocorticoid treatment.

Limitations of the study

Several important questions arise from our work. For instance, accumulated senescent BMA δ s have been observed in old mice,⁴⁸ but is oxylipin-PPAR γ -INK signaling also involved in the senescence of BMA δ s during aging? Bone-marrow adipose tissue mass increases under many pathological conditions, such as obesity, type 2 diabetes, anorexia, and hyperlipidemia. Is there senescent BMA δ accumulation under these conditions? Does the identified mechanism, i.e., oxylipin-PPAR γ -INK signaling, also mediate BMA δ senescence in these conditions? Is BMA δ senescence a key contributor to bone-marrow adiposity? Finally, we are aware that it is technically challenging to fully exclude the influence of peripheral adipose tissue on glucocorticoid-induced bone remodeling changes. It is important to determine whether glucocorticoids also induce cellular senescence in peripheral adipose tissue and whether transplanting senescent peripheral adipose tissue from mice may have similar adverse bone effects. These experiments are important to define the distinct role of BMA δ s and peripheral adipose tissue in regulating bone and BM in pathological settings. Recently, a BMA δ -specific Cre mouse model has been generated.⁷⁶ This mouse line serves as a useful tool to study the specific contribution of BMA δ s to glucocorticoid-induced bone deterioration.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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- RESOURCE AVAILABILITY

BMA δ s per mm² tissue area (N. β Gal⁺ BMA δ s) (B). Representative micro-CT images of p16 cKO or WT littermates treated with vehicle or DEX are shown in (C) with quantitative analysis of trabecular bone volume/tissue volume (BV/TV) (D), trabecular thickness (Tb. Th) (E), trabecular number (Tb. N) (F), and trabecular separation (Tb. Sp) (G).

(H) Schematic diagram showing the experimental design for transplantation.

(I–M) Representative micro-CT images of WT mice transplanted with BMA δ s from p16 cKO or WT littermates treated with vehicle or DEX are shown in (I) with quantitative analysis of BV/TV (J), Tb. Th (K), Tb. N (L), and Tb. Sp (M).

(N) Diagram illustrating the mechanisms of glucocorticoid-induced BMA δ senescence in bone and bone marrow. (n = 5 mice.) Data are represented as mean $\pm$ SEM. *p < 0.05, **p < 0.01, ***p < 0.001 as determined by two-way ANOVA.

- Lead contact
- Materials availability
- Data and code availability
- **EXPERIMENTAL MODEL AND SUBJECT DETAILS**
 - Mouse models
 - Cell lines
- **METHOD DETAILS**
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 - Senescence-associated distension of satellites (SADS) analysis
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 - RNA-sequencing data analysis
- **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cmet.2023.03.005>.

ACKNOWLEDGMENTS

The authors acknowledge the assistance of Rachel Box, MS, at the Johns Hopkins Department of Orthopaedic Surgery Editorial Services for editing the manuscript. We thank Judith Campisi (Buck Institute for Research on Aging) and Gloria H. Su (Columbia University Medical Center) for kindly providing the *p16-3MR* mice and *p16^{fl/fl}* mice, respectively. This work was supported by the National Institutes of Health grant R01AG068226 and R01AG072090 to M.W., and P01AG066603 to X.C.

AUTHOR CONTRIBUTIONS

X.L. and M.W. designed the experiments, analyzed results, and wrote the manuscript. X.L. carried out most of the experiments. Y.G., S.K., S.A., J.W., and C.-L.F. helped with some experiments. M.W. supervised the experiments. X.L. drew the illustrations in the figures. X.C. proofread the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We worked to ensure sex balance in the selection of non-human subjects. One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in their field of research or within their geographical location. One or more of the authors of this paper self-identifies as a gender minority in their field of research. While citing references scientifically relevant for this work, we also actively worked to promote gender balance in our reference list.

Received: June 9, 2022
Revised: January 12, 2023
Accepted: March 6, 2023
Published: April 4, 2023

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rat monoclonal anti-Emcn	Santa-Cruz	CAT# sc-65495; RRID:AB_2100037
Mouse/Rat polyclonal anti-CD31	R&D Systems	CAT# FAB3628G; RRID:AB_10972784
Rabbit monoclonal anti- γ H2AX	Cell signaling Technology	Cat# 9718; RRID:AB_2118009
Rabbit polyclonal anti-Perilipin	Sigma	CAT# P1873; RRID:AB_2118009
Rabbit polyclonal anti-Perilipin	Cell signaling Technology	CAT# 9349; RRID:AB_10829911
Rabbit polyclonal anti-Osterix	Abcam	CAT# ab22552;RRID:AB_2194492
Mouse monoclonal anti-Osteocalcin	Takara	CAT# M188; RRID: AB_2935810
Rabbit polyclonal anti-RFP	Rockland Immunochemicals	CAT# 600-401-379;RRID:AB_2209751
Rabbit polyclonal to HMGB1	Abcam	CAT# ab18256; RRID:AB_444360
Rat monoclonal anti-CD144	Biolegend	CAT# 138013; RRID:AB_2562627
Rat monoclonal anti-CD45	Biolegend	CAT# 103111; RRID:AB_312976
Mouse monoclonal anti-CD31	Biolegend	CAT# 303115; RRID:AB_1877152
Rat monoclonal anti-Sca1	Biolengend	CAT# 108125; RRID:AB_10639725
Rat monoclonal anti-CD24	Biolegend	CAT# 101825; RRID:AB_10901159
Mouse monoclonal anti-Lamin-B1	Santa cruz	CAT# sc-374015; RRID:AB_10947408
Rabbit monoclonal anti-human p16	Abcam	CAT# 211542; RRID:AB_2891084
Rabbit monoclonal anti-mouse p16	Cell signaling Technology	CAT# 80772; RRID:AB_2799960
Rabbit monoclonal anti-mouse p21	Cell signaling Technology	CAT# 37543; RRID: AB_2935811
Rabbit monoclonal anti-human p21	Abcam	CAT# ab109520; RRID:AB_10860537
Rabbit monoclonal anti-GAPDH	Cell Signaling	CAT# 2118; RRID:AB_561053
Chemicals, peptides, and recombinant proteins		
Collagenase A	Sigma	CAT# 11088793001
Trypsin	Gibco	CAT# 27250018
Ganciclovir	MilliporeSigma	CAT# 82410-32-0
Rosiglitazone	MilliporeSigma	CAT# 155141-29-0
T0070907	MilliporeSigma	CAT# 313516-66-4
GW9662	MilliporeSigma	CAT# 22978-25-2
Dasatibib	MilliporeSigma	CAT# 302962-49-8
Quercitin	MilliporeSigma	CAS# 117-39-5
15d-PGJ2	MilliporeSigma	CAS# 87893-55-8
Celecoxib	MilliporeSigma	CAS# PZ008
Critical commercial assays		
Senescence β Gal staining kit	Cell Signaling Technology	Cat# 9860S
Mouse XL cytokine array kit	R&D System	Cat# ARY028
PGD2-MOX ELISA kit	Cayman Chemical Company	Cat# 500151
PGJ2 ELISA kit	ENZO	Cat# ADI-900-023
PGE2 ELISA kit	R&D System	Cat# KGE004B
RNeasy Mini Kit	QIAGEN	Cat# 74014
BCA protein assay kits	Thermo Fisher Scientific	Cat# 23227
Deposited data		
RNA-Seq of differentiated BMAds	GEO Database	GSE205732
Data S1 . Unprocessed source data underlying all blots and graphs.	This paper	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: Cell lines		
Human umbilical vein endothelial cells(HUVEC)	ATCC	Cat# PCS-100-013
Experimental models: Organisms/strains		
Mouse: AdipoQ-Cre mice	The Jackson Laboratory	JAX: 010803
Mouse: C57BL/6 floxed <i>p16</i>	Qiu et al. ⁷⁷	https://www.entcolumbia.org/research/research-labs/su-lab
Mouse: p16-tdTomreporter mice (C57BL/6 background)	Liu et al.	https://doi.org/10.1073/pnas.1818313116
Mouse: C57BL/6J	The Jackson Laboratory	JAX:000664
Oligonucleotides		
PCR primer for genotype: Adipoq-Cre forward: 5'- GGA TGT GCC ATG TGA GTC TG -3'	The Jackson Laboratory	https://www.jax.org/Protocol?stockNumber=010803&protocolID=25488
PCR primer for genotype: Adipoq-Cre reverse: 5'- ACG GAC AGA AGC ATT TTC CA-3'	The Jackson Laboratory	https://www.jax.org/Protocol?stockNumber=010803&protocolID=25488
PCR primer for genotype: p16-floxed forward: 5'- AGG AGT CCT GGC CCT AGA AA-3'	Qiu et al.	https://doi.org/10.18632/oncotarget.357
PCR primer for genotype: p16-floxed reverse: 5'- CCA AAG GCA AAC TTC TCA GC-3'	Qiu et al.	https://doi.org/10.18632/oncotarget.357
RT-PCR primer: Mki67 forward: 5'-ACCGTGGAGTAGTTTATCTGGG-3'	Liu et al.	N/A
RT-PCR primer: Mki67 Reverse: 5'-TGTTTCCAGTCCGCTTACTTCT-3'	Liu et al.	N/A
RT-PCR primer: Cdkn2a Forward: 5'- GAAAGAGTTCCGGGCGTTG-3'	Liu et al.	N/A
RT-PCR primer: Cdkn2a Reverse 5'-GAGAGCCATCTGGAGCAGCAT-3'	Liu et al.	N/A
RT-PCR primer: Tp53 Forward: 5'-ATCGCCTTCGACATCATCGC-3'	Liu et al.	N/A
RT-PCR primer: Tp53 Reverse: 5'-CCCCATGCGTACTCCATGAG-3'	Liu et al.	N/A
RT-PCR primer: Cdkn1a Forward: 5'-AGAAGGTACTTACGGTGTGGT-3'	Liu et al.	N/A
RT-PCR primer: Cdkn1a Reverse: 5'-GAGAGATTTCCCGAATTGCAGT-3'	Liu et al.	N/A
RT-PCR primer: COX-2 Forward: 5'- CCACTTCAAGGGAGTCTGGA-3'	This paper	N/A
RT-PCR primer: COX-2 Reverse: 5'- AGTCATCTGCTACGGGAGGA-3'	This paper	N/A
RT-PCR primer: Pparg Forward: 5'- GTACTGTGCGTTTCAGAAGTGCC-3'	This paper	N/A
RT-PCR primer: Pparg Reverse: 5'- ATCTCCGCCAACAGCTTCTCCT-3'	This paper	N/A
Software and algorithms		
NRecon image reconstruction software	Bruker MicroCT	Version 1.6
CTAn data-analysis software	Bruker MicroCT	Version 1.9
CTVol 3-dimensional model visualization software	Bruker MicroCT	Version 2.0
Prism 9	Graphpad software	http://www.graphpad.com/
Flowjo_v10.8	Flowjo LLC.	http://www.flowjo.com/
Zen black	Carl Zeiss Microscopy	https://www.zeiss.com/microscopy/
Other		
Vascular Cell Basal Medium	ATCC	Cat# PCS-100-030
Endothelia Cell Growth Kit-BBE	ATCC	Cat# PCS-100-040

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
MesenCult MSC Expansion Kit	Stem Cell Technologies	Cat# 05513
StemPro Adipogenesis Differentiation kit	Gibco	Cat# A1007001

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Mei Wan (mwan4@jhmi.edu).

Materials availability

- This study did not generate new unique reagents.

Data and code availability

- RNA-Seq of differentiated BMAds data has been deposited in the NCBI GEO Database and is publicly available as of the date of publication with the accession number: GEO: GSE205732.
- Unprocessed source data underlying all blots and graphs is deposited in [Data S1](#).
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Mouse models

AdipoQ-Cre mice (C57BL/6 background, stock no. 010803) were purchased from The Jackson Laboratory (Bar Harbor, ME). *p16^{flox/flox}* mice (C57BL/6 background) were generated by Dr. Gloria H. Su's laboratory from the Department of Pathology, Columbia University Medical Center.⁷⁷ *p16^{tdTom}* reporter mice (C57BL/6 background) were generated by Dr. Norman E. Sharpless's laboratory, University of North Carolina School of Medicine (Chapel Hill, NC).⁴⁷

We crossed the 3-month-old *AdipoQ-Cre* mice with 3-month-old *p16^{flox/flox}* mice. The offspring were backcrossed with *p16^{flox/flox}* mice (both sexes) to generate *AdipoQ-Cre::p16^{flox/flox}* (*p16 cKO*) and *p16^{flox/flox}* (*WT*) mice. 3-month-old male *p16 cKO* mice and *WT* mice were used for experiments. The genotypes of the mice were determined by PCR analyses of genomic DNA extracted from mouse-tail snips using the following primers: *AdipoQ-Cre* forward, 5'-GGA TGT GCC ATG TGA GTC TG-3' and reverse, 5'-ACG GAC AGA AGC ATT TTC CA -3'; loxP *p16* allele forward, 5'-AGG AGT CCT GGC CCT AGA AA-3' and reverse, 5'-CCA AAG GCA AAC TTC TCA GC-3'.

To detect bone phenotypes after glucocorticoid treatment, we injected 3-month-old male C57BL/6 mice with MPS (10 mg/m²/day) for different time periods (3 days, or 1, 2 or 4 weeks). Body surface area (BSA) was calculated using Meeh's formula, with a k constant of 9.82 for mice × body weight (g) to the two-thirds power (BSA = kW^{2/3}). For ruxolitinib (JAKi) treatment, JAKi was dissolved in 10% DMSO and administered by oral gavage at a dosage of 60 mg/kg daily for 2 weeks as previously described.⁴⁸

All mice were maintained at the animal facility of The Johns Hopkins University School of Medicine (Baltimore, Maryland) under protocol MO18M139 approved by the Institutional Animal Care and Use Committee of The Johns Hopkins University. Mice were housed in normal conditions with a 12h:12h light:dark cycle in a temperature-controlled room with food and water provided *ad libitum*.

Cell lines

In this study, human umbilical vein endothelial cells (HUVECs) were purchased from ATCC (Cat. No, PCS-100-013). Cells were cultured in Vascular Cell Basal Medium (PCS-100-030, ATCC) supplemented with Endothelial Cell Growth Kit-BBE (PCS-100-040, ATCC) and penicillin/streptomycin at 37°C.

Primary bone marrow stromal cells (BMSCs), BMAds and bone marrow adipocyte precursor cells (APC) were isolated from 3-month-old C57BL/6 mice. BMSCs were maintained in BMSCs growth media (MesenCult MSC Expansion Kit, Stem Cell Technologies, Cat #05513) with penicillin/streptomycin at 37°C. BMAds and adipocyte precursor cells (APC) were maintained in DMEM supplemented with 15% fetal bovine serum and penicillin/streptomycin at 37°C. Primary osteoblasts were isolated from young (2-week-old) C57BL/6 mice and maintained in DMEM supplemented with 15% fetal bovine serum and penicillin/streptomycin at 37°C. For primary cell isolation protocols see "method details" section.

METHOD DETAILS

μ CT analysis of the femoral bone

Femurs were harvested from mice, dissected free of soft tissue, and fixed in 4% phosphate-buffered paraformaldehyde for 48 hours. For μ CT analysis, mice femora were dissected free of soft tissue, fixed overnight in 10% formalin at 4°C and analyzed by high-resolution μ CT (Skyscan 1172, Bruker MicroCT, Kontich, Belgium). The scanner was set at 65 kV, 153 μ A and a resolution of 9.0 μ m/pixel. We used NRecon image reconstruction software, version 1.6 (Bruker MicroCT), CTAn data-analysis software, version 1.9 (Bruker MicroCT), and CTVol 3-dimensional model visualization software, version 2.0 (Bruker MicroCT) to analyze parameters of the trabecular bone in the metaphysis. To perform 3-dimensional histomorphometry analysis of trabecular bone, we selected the regions of interest from 1mm below the distal epiphyseal growth plate(reference level) and extending toward the distal direction for proximally 2mm in length. Trabecular bone was analyzed to determine trabecular BV/TV, Tb.Th, Tb.N, and Tb.Sp. Femur length was analyzed using CTAn data-analysis software, version 1.9 (Bruker MicroCT).

Histological analysis

Femoral bones were dissected and freshly fixed in 4% paraformaldehyde overnight, followed by a 14-day decalcification in 0.5M EDTA (pH 7.4). The samples were then dehydrated in 20% sucrose plus 2% polyvinylpyrrolidone solution for 24 hours and embedded in OCT. Ten μ m-thick coronal sections of the femurs were obtained for SA- β Gal staining. Senescent cells were detected using a senescence β Gal staining kit according to the manufacturer's instructions (Cell Signaling Technology, Danvers, MA). For Oil Red O staining following SA- β Gal staining, the slides were rinsed in PBS for 3 times, placed in 100% propylene glycol for 2 min, and stained in filtered 0.5% Oil Red O solution in propylene glycol for 30 min. The slides were transferred into an 85% propylene glycol solution for 1 min, rinsed in PBS 3 times, mounted, and sealed with nail polish. Representative images were acquired with a BX51 microscope (Olympus, Tokyo, Japan).

Twenty μ m-thick coronal sections of the femurs were obtained for immunofluorescence staining. Bone sections were blocked in PBS with 3% BSA for 1 hour and then stained overnight (>8 hours) with a primary antibody: Emcn (Santa-Cruz, sc-65495, 1:50), CD31 (R&D Systems, FAB3628G, 1:100), γ H2AX (Cell signaling, 20E3, 1:200), Perilipin (Sigma, P1873, 1:500), Perilipin (Cell signaling, 9349, 1:200), Osterix (Abcam, ab22552, 1:200), Osteocalcin (Takara, M188, 1:200), RFP (Rockland Immunochemicals, 600-401-379, 1:200) and HMGB1 (Abcam, ab18256, 1:300). Fluorescence-conjugated secondary antibodies (Jackson ImmunoResearch, 1:200) were used in immunofluorescence procedures to detect fluorescent signals. Slides were then mounted with anti-fade prolong gold (Invitrogen) mounting medium and sealed with nail polish, after which representative images were acquired using a Zeiss LSM780 confocal microscope or an Olympus BX51 microscope.

Senescence-associated distension of satellites (SADS) analysis

SADS assay, which is a fluorescence in situ hybridization (FISH) for detection of peri-centromeric satellite heterochromatin DNA, was performed following the protocol kindly provided by Drs. Sundeep Khosla and Joshua Farr.⁴⁸ Briefly, paraffin-embedded femoral bone tissue sections were cross-linked with 4% paraformaldehyde and dehydrated in ethanol. The bone tissue sections were then denatured and hybridized in the buffer containing 1.0 μ g/mL Cy3-labeled, CENPB-specific (ATTCGTTGGAAACGGGA) peptide nucleic acid FISH probe (Panagene Inc, Korea) for 2 hours at room temperature in the dark. After washing, immunofluorescence staining of the tissue sections was performed using primary antibody against perilipin (Novus Biologicals, NB100-60554, 1:500). Finally, the slides were mounted with DAPI-containing mounting media (Life Technologies). Images were acquired with a Zeiss LSM780 confocal microscope, and the intensity profiles were acquired with Image J software (NIH).

FACS sorting and flow cytometry

Flow cytometric analysis and FACS sorting of bone vascular cells from femur were performed as previously described^{78,79} with modifications. Briefly, the epiphysis was removed from the distal femora and proximal tibia, and the trabecular bones were then crushed in ice-cold PBS with a mortar and pestle. Whole bone marrow was digested with collagenase A (Sigma, 11088793001, 2 mg/mL) and trypsin (Gibco, 27250018, 2.5 mg/mL) in PBS at 37°C for 30 min to obtain single-cell suspensions. Ammonium-chloride-potassium lysing buffer was used to remove red blood cells. For sorting of vascular cells, after washing and filtration, cells were incubated for 45 min at 4°C with CD144 antibody (BV421, Biolegend, 1:50) and sorted using BD FACS Aria II (BD Biosciences). *p16^{tdTom}* mice were used for the analyses of the incidence of tdTom⁺ cells in different bone marrow cell populations. The antibody used in this study to identify vascular endothelial cells was anti-CD144 (BV421, Biolegend, 1:50). The antibodies used to identify BMSCs and APCs were anti-CD45 (APC, Biolegend, 1:50), anti-CD31 (APC, Biolegend, 1:50), anti-Sca1 (APC-Cy7, Biolegend, 1:50) and anti-CD24 (BV421, Biolegend, 1:50). Flow cytometry cells were analyzed using a BD LSRII flow cytometer.

Isolation of BMAdS and adipocyte precursor cells (APC)

Mature BMAdS were isolated directly from bone marrow of mice according to previously described protocols.^{80,81} Briefly, femurs and tibias were collected from mice, and the two ends of the bones were snapped. The bones were placed in a small microcentrifuge tube (0.6 mL) that was cut open at the bottom. The small tube with the bones were then placed into a bigger microcentrifuge tube (1.5 mL). Fresh bone marrow was spun out by quick centrifuge. Red blood cells were lysed using RBC lysing buffer (Sigma). After centrifugation (3000 rpm, 5 minutes, RT), floating adipocytes were collected from the top layer and washed with PBS for 3 times. To isolate

APCs, cell pellet from the previous step was collected and washed with PBS for 3 times. The cells were then resuspended and incubated with CD45 (APC, Biolegend, 1:50), CD31 (APC, Biolegend, 1:50), Sca1 (APC-Cy7, Biolegend, 1:50) and CD24 (BV421, Biolegend, 1:50). FACS sorting of CD45⁺CD31⁺CD24⁺Sca1⁺ APCs was performed using BD FACS Aria II (BD Biosciences).

Cell transplantation

To induce senescence of BMAds to be transplanted, donor mice were injected with MPS (10 mg/m²/day) for 2 weeks. BMAds and APCs were isolated, respectively as described in the above section. For intra-femoral transplantation, acceptor mice (C57BL/6) were anesthetized, and a longitudinal skin incision was made across the front of the right knee to visualize patellar tendon. A 27G needle was inserted through the patellar tendon, between the femur condyles, and the needle was advanced slowly in a twisting motion through the bone surface to 2–4mm deep. Successful penetration was confirmed by slow but constant bleeding. The cell suspensions of BMAds (3×10³) or APCs (2×10⁴) in 20 μl PBS were slowly injected into the medullary space of the femur. The hole was immediately sealed with bone wax, and the skin was sutured.

BMSCs isolation, culture, and adipogenesis differentiation

BMSCs isolation, culture, and adipogenesis differentiation were conducted according to previously described protocol with customization.^{82,83} Briefly, 3-month-old male C57BL/6 mice were euthanized, and femurs and tibias were dissected. Bone marrow cells were flushed by using 22-gauge syringes with PBS. The freshly isolated single-cell suspensions were plated at a density of 5 × 10⁴ /cm² in 6-well plates in BMSCs growth media (MesenCult MSC Expansion Kit, Stem Cell Technologies, Cat #05513). Cells were allowed to grow for 3 days before supernatant was aspirated, rinsed with PBS 3 times, and medium was changed 3 times a week. For adipogenesis differentiation, MSCs were cultured with adipogenesis media (StemPro Adipogenesis Differentiation kit, Gibco) for designated days with medium changed 3 times a week.

Primary osteoblast isolation

Primary osteoblasts were isolated from young (2-week-old) mice. Briefly, calvarium was aseptically dissected and processed by serial digestion using 300 active U/mL of collagenase type-1A (Sigma-Aldrich) dissolved in α -minimal essential medium (α MEM; Gibco). All steps of the digestion took place in a 6-well Petri dish, on a rotating shaker set to 200 RPM, in a 37°C and 5% CO₂ humidified incubator. Following each sequential digestion, the digest solution with suspended cells was removed from the bone fragments and kept. The bone fragments were rinsed with Hank's balanced salt solution 3 times, and the rinse buffer was added to the digestion solution. The combined cell suspension solution was spun down at 200 × g for 5 min, followed by removal of supernatant from the cell pellet and resuspension of the cells in culture medium.

In vitro cell treatment and staining

For treatments, 10^{−6} M DEX was added to cells and cultured for designated times. Other drugs and compounds used in this study were as follows: ganciclovir (GCV, MilliporeSigma, 20×10^{−6} M), rosiglitazone (MilliporeSigma, 20×10^{−6} M), T0070907 (MilliporeSigma, 10^{−6} M), GW9662 (MilliporeSigma, 10^{−6} M), dasatinib (MilliporeSigma, 10^{−6} M) + quercetin (MilliporeSigma, 10×10^{−6} M), 15d-PGJ2 (MilliporeSigma, 10×10^{−6} M), and celecoxib (MilliporeSigma, 50×10^{−6} M). Dosages and time courses are noted in the corresponding text and figure legends.

For SA- β Gal staining, cells were fixed and stained with a senescence β Gal staining kit according to the manufacturer's instructions (Cell Biolabs). For immunofluorescence staining, cells were washed with PBS, fixed in 10% formalin, and blocked with 30 mg/ml BSA/PBS at 37°C for 1 hour. Cells were stained with primary antibody at 4°C overnight. Primary antibody used were: γ H2AX (Cell signaling, 20E3, 1:200), Perilipin (Cell signaling, 9349, 1:200) and Lamin-B1 (Santa cruz, sc-374015, 1:200). Fluorescence-conjugated secondary antibodies (Jackson ImmunoResearch, 1:200) were used in immunofluorescence procedures to detect fluorescent signals. Cells were then mounted with anti-fade prolong gold mounting medium (Invitrogen) and imaged with inverted fluorescence microscope (Olympus).

Proteomic profiling in conditioned medium (CM) collected from cell culture

CM was collected from each group after centrifugation at 700 × g for 10 min at 4°C and stored at −80°C for subsequent experiments. For cytokine measurement, conditioned medium was subjected to a proteome profiler array using a mouse XL cytokine array kit (R&D, ARY028), and the relative levels of 111 cytokines and chemokines were measured. The array procedure and data analysis were performed according to the manufacturer's instructions.

For measurement of prostaglandins, CM were collected as described and analyzed according to the manufacturer's instructions. The ELISA kit used for measurement were as follows: PGD2-MOX ELISA (Cayman Chemical company, 500151); PGJ2 ELISA (ENZO, ADI-900-023); PGE2ELISA (R&D systems, KGE004B).

Western blot analysis

Western blot analysis was conducted as previously described.^{84,85} Briefly, cells were lysed in RIPA lysis buffer and harvested with a rubber policeman. Samples (20 μg protein) were loaded into pre-cast electrophoresis gels (Bio-Rad), separated by SDS-PAGE and electro-transferred onto a PVDF membrane. After blocking with 5% nonfat milk, the blots were incubated with primary antibodies overnight at 4°C and then with secondary antibodies at room temperature for 1 hour. Blots were developed with the SuperSignal

West Femto chemiluminescence kit (Thermo Scientific) followed by autoradiography. The following antibodies were used: anti-mouse p16 (Abcam, 211542, 1:1000), anti-human p16 (Cell Signaling, 80772, 1:1000), anti-mouse p21 (Cell Signaling, 37543, 1:1000), anti-human p21 (Abcam, ab109520, 1:1000), and GAPDH (Cell Signaling, 2118, 1:1000).

qRT-PCR

Total RNA for qRT-PCR was purified from the cultured or sorting cells using RNeasy Mini Kit (QIAGEN) according to the manufacturer's protocol. Complementary DNA (cDNA) was prepared with random primers using the SuperScript First-Strand Synthesis System (Invitrogen) and analyzed with SYBR GreenMaster Mix (QIAGEN) in the thermal cycler with 2 sets of primers specific for each targeted gene. Target-gene expression was normalized to glyceraldehyde 3-phosphate dehydrogenase (GAPDH) messenger RNA, and relative gene expression was assessed using the $2^{-\Delta\Delta CT}$ method. Primers used for qRT-PCR were as follows: *Mki67* (5'-ACCGTGGAGTAGTTTATCTGGG-3') and (5'-TGTTTCCAGTCCGCTTACTTCT-3'); *Cdkn2a* (5'-GAAAGAGTTCGGGGCGTTG-3') and (5'-GAGAGCCATCTGGAGCAGCAT-3'); *Tp53* (5'-ATCGCCTTCGACATCATCGC-3') and (5'-CCCCATGCGTACTCCATGAG-3'); *Cdkn1a* (5'-AGAAGGTACTTACGGTGTGGT-3') and (5'-GAGAGATTCCCGAATTGCAGT-3'); *Cox-2* (5'-CCACTTCAAGGGAGTCTGGA-3') and (5'-AGTCATCTGCTACGGGAGGA-3'); and *Pparg* 5'-GTACTGTCTGGTTTCAGAAGTGCC-3') and (5'-ATCTCCGCCAACAGCTTCTCCT-3').

Preparation and sequencing of RNA-sequencing libraries

Bone marrow adipocytes were differentiated from isolated bone marrow stem cells with StemPro Adipogenesis Differentiation kit (Gibco) for 14 days and treated with 10^{-6} M DEX or 10^{-6} M DEX plus GW9662 (MilliporeSigma, 10^{-6} M) for 24 hours. The cells were lysed and total RNA was collected using RNeasy Mini Kit (QIAGEN, 74014) following manufacturer's instruction. RNA integrity was assessed using Agilent RNA 6000 Nano kit (Agilent), and RNA with a RIN > 7 was used in sequencing. Libraries were prepared using the Smart-Seq 2 protocol with a template switching oligonucleotide, followed by PCR amplification and purification. The constructed cDNA libraries were sequenced using an Illumina HiSeq 2000. The RNA-seq data have been deposited to the GEO repository (GSE205732).

RNA-sequencing data analysis

Sample demultiplexing and conversion to FASTQ files was performed using Illumina's fastq software with all default options. Reads from FASTQ files of the cultured cells were aligned to mouse genome using the STAR aligner (v2.4.2). Uniquely mapped reads were used for gene expression estimates as transcripts per million reads (TPMs). Kruskal-Wallis tests were used for identifying genes differentially expressed among sample groups. Pathway analysis was performed for functional annotation of the ASIGs in the dataset using established tools available online (EnrichR). Heatmap was formed using established online tool (Bioinfo intelligent cloud).

QUANTIFICATION AND STATISTICAL ANALYSIS

Data are presented as means $\pm$ standard errors of the mean. Unpaired, 2-tailed Student *t*-tests were used for comparisons between 2 groups. For multiple comparisons, 1-way or 2-way analysis of variance with Bonferroni post hoc test was used. All data were normally distributed and had similar variation between groups. Statistical analysis was performed using SPSS software, version 15.0 (IBM, Armonk, NY). $p < 0.05$ was deemed significant. All representative images of bones or cells were selected from at least 3 independent experiments with similar results unless indicated otherwise in the figure legend.